

Tactical choice in repeated protests

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Abstract

Why do some protest movements escalate from peaceful demonstrations to violent confrontation while others never change tactics? I develop a dynamic model in which an organizer chooses between tactics that differ in how they convert participation into political power. Failed protests reveal information about regime strength, reducing subsequent mobilization. Optimal tactical sequencing reflects not only how powerful each tactic is today, but how valuable it may be in a lower-mobilization environment tomorrow. I show that movements preserve tactics that remain effective after demobilization, generating path-dependent patterns of tactical persistence or learning-motivating switching. Extending the model to repression, I show that state intervention can redirect movements across tactics and, by changing the informational content of failures, partly or even fully undermine the benefits of repression. As a technical contribution, I introduce a cost-signals structure which resolves equilibrium multiplicity in a dynamic global game with public learning.

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1 Introduction

What determines how protest movements choose and change tactics over time? Some movements undergo striking transformations in their main mode of contestation. For instance, the African National Congress turned from civil disobedience to violent sabotage and insurgency with the creation of Umkhonto we Sizwe; the initial peaceful street gatherings which characterized the Arab Spring later gave way to violent clashes and armed confrontations; climate protests have pivoted from large-scale marches towards more radical tactics of targeted disruption to infrastructure and everyday life. Other movements never transform at all: Poland's Solidarity movement remained overwhelmingly nonviolent and mass-based throughout the 1980s, while a century earlier, the January Uprising of 1863 took the form of armed insurrection from the start. What explains these divergent trajectories?

Canonical theoretical accounts emphasize how nascent protest movements grow by exposing latent sympathies, lowering barriers to participation, and generating mobilization cascades (Kuran 1991; Lohmann 1994). However, the same informational feedback loop that helped to initially launch a protest can quickly work in reverse as failed protest episodes accumulate. When the movement repeatedly falls short of its objective, citizens gather evidence that the regime is stronger, and the movement's prospects worse than they previously hoped. Failure therefore erodes confidence in subsequent attempts and, with it, the participation on which many forms of contention depend. Therefore, while existing accounts tell us a great deal about how information causes movements to grow or peter out, a key open question is how movements *change what they do* as public confidence deteriorates.

This paper argues that tactical choice plays a central role in how movements navigate and manage declines in public confidence. Different tactics convert participation into political power in different ways. Mass demonstrations and occupation-style tactics generate extraordinary leverage when participation is broad, but lose much of their force when turnout falls. By comparison, violent or disruptive tactics tend to retain a substantial amount of power with relatively few participants, but benefit relatively less from additional participation. These technological differences interact with variation in the movement itself: When potential participants respond similarly to changes in confidence, even a modest deterioration in beliefs can produce a sharp decline in turnout. By contrast, when a movement is more stratified and participation propensities are more dispersed, mobilization is comparatively resilient. The sensitivity of participation to beliefs thus shapes the severity of the dynamic problem the movement faces.

At the heart of the argument is the feedback between information, mobilization, and tactical effectiveness. A protest outcome changes beliefs about regime strength, which affect subsequent participation. Changes in participation then alter the relative productivity of mass-based versus disruptive tactics. Hence, past protests change the environment in which *future* tacti-

cal decisions may need to be made. Forward-looking movement leaders who determine protest strategy must therefore balance two considerations: First, which tactic is most likely to achieve success in the present, and second, which tactics might be most valuable if saved for a future state created by failure. The problem resembles that of dynamic experimentation, with the additional feature that what an action reveals then changes the future effectiveness of each available action.

I develop this argument in a two-period model of repeated protest. In each period, two stylized protest technologies are available: a peaceful mass tactic and a violent or disruptive tactic. Peaceful power rises strongly with participation and can achieve the highest possible level of pressure if mobilization is full. Violent power has a guaranteed floor of power when turnout is low, but its responsiveness to additional participation can vary; it generates less leverage than peaceful power at full mobilization. An organizer (such as a movement leader or coordinating body) chooses tactics to maximize the probability that the movement eventually succeeds; this abstraction captures coordinated decisions over strategy even though leaders cannot compel individual citizens to participate. Citizens decide individually whether to participate based on common beliefs about regime strength, and privately realized participation costs. A failed protest revises citizens' beliefs towards stronger regime types, eating away at expected participation in the next attempt. How much turnout responds to beliefs on the dispersion of participation propensities. The model thus separates technological features – the advantage of violence at lower turnout, and the advantage of peace at higher turnout – from the sensitivity of participation to changing confidence, a feature of movement composition.

The first result shows how movements trade off exploiting favorable conditions today against preserving robust tactics for tomorrow. A stark benchmark isolates the dynamic value of preserving a robust tactic: Suppose violence is completely insensitive to turnout, analogous to a form of pre-positioned sabotage requiring little or no direct human presence to trigger. Even if this tactic provides substantial guaranteed power, the organizer never uses it first. Precisely because its effectiveness does not depend on public confidence, postponing it carries no cost: it will remain equally useful after a failed first attempt. Peaceful mass mobilization, by contrast, is “perishable.” Since it is most effective while confidence and expected turnout remain high, the organizer wants to use this tactic first – *even if its expected power is lower than that of violence* – and holds the robust technology in reserve.

The benchmark characterizes the dynamic incentive to preserve violence for the future. However, it does not capture the fact that some forms of violence, however, may still substantially benefit from favorable mobilization – for instance, riots and armed confrontations in urban centers. This generates an opposing *technological* incentive to use violence immediately. Which of the two forces dominates depends on the resilience of mobilization to public beliefs.

When mobilization is relatively resilient, sufficiently scalable violence is worth using early on, while turnout is strong. As mobilization levels become more sensitive to confidence, the balance becomes dominated by the dynamic preservation motive. When sensitivity is extremely high, this motive dominates even when violence is arbitrarily close to the highly scalable peaceful tactic. High dependence on public confidence therefore determines whether violence is worth exploiting while mobilization is favorable or preserving as insurance against a pessimistic future.

The second set of results concerns how movements adapt to failures in the baseline model. Contrasting existing explanations which attribute violent turns to radicalization or desperation, the model identifies an informational explanation. Large, failed protests have two competing consequences: They are highly damaging to beliefs and turnout, seeming to favor the violent tactic. But a *large* failure is suggestive that the surviving regime is strong – and that the lower ceiling of the violent tactic may be ineffective. As I show, this second selection effect dominates in the model. Relatively modest failures, which are not very informative about the regime’s strength, leave violence attractive as a fallback that insures at least some level of baseline capacity. After sufficiently strong failures, by contrast, the organizer returns to the tactic capable of generating the greatest possible power – peaceful mass mobilization. When movements turn to violence in the second period, it is not because the first failure is devastating, but because it is actually relatively uninformative.

These results give structural features of contentious politics a dynamic interpretation. Terrain, infrastructure, organizational capacity, communications technology, population density, and access to politically important public space all affect the kinds of tactics available to protests (Beissinger 2022; Tarrow 1998; Tilly 1978). The model distinguishes two ways in which such conditions can strengthen violent contention. Some features – for instance, rugged terrain or vulnerable infrastructure – can bolster the amount of power that very small groups can generate. Other features can make violence more similar to a mass tactic: urban density, for instance, can increase the importance of mass participation to confrontations with armed forces or attempts to seize control of property. These two types of violent capacity have different dynamic consequences: higher baseline capacity makes violence valuable as insurance for the future, while higher scalability makes violence more attractive as an opening tactic. Structural conditions thus determine the tactics a movement possesses *as well as* the incentives that ultimately decide how those tactics are or are not used.

This perspective also clarifies why movements with similar repertoires but different mobilization bases may follow different tactical trajectories. A movement with a committed core whose participation is resilient to past setbacks has less reason to fear the effects of pessimism and can afford to use violence aggressively while conditions are favorable. A movement whose participation is fragile – swinging from large to negligible – as confidence deteriorates – instead

places greater value on holding violence in reserve for pessimistic future states.

The same feedback between information, mobilization, and tactical effectiveness bears implications for repression. Regimes frequently intervene in the tactical environment: policing, restricting access to public space, and investments in security measures all alter the relative effectiveness of different parts of a movement’s repertoire. However, because movements use their repertoire to strategically adjust to repression, repression reshapes the form and trajectory of protest movements, and may not always successfully tamp down contention. I extend the model with a regime actor who can choose to use (costly) repression against tactics to minimize the protest’s chance of success. The regime’s problem mirrors that of the organizer. Weakening a tactic can improve the regime’s prospects of survival today, but the intervention also changes what citizens learn from failure and therefore the tactical environment the movement inherits tomorrow.

I demonstrate two limits to the efficacy of state intervention. The first comes from *tactical substitution*. Repressing a tactic can prompt the organizer to switch to the alternative tactic once it becomes sufficiently attractive. State intervention may then redirect contention rather than simply suppress it. The key implication of this is that repression can reverse the learning-driven patterns of tactical adjustment described in the baseline model. Following a weak failure, repressive measures to wipe out an insurgency can redirect a movement that would otherwise turn to violence back toward a mass peaceful march; following a stronger failure, repressing peaceful protests that would otherwise occur could make violence attractive. The efficacy of the unused alternative tactic therefore limits the value that repression can extract, even when it is never observed in equilibrium.

The second constraint is informational. Temporary repression may help the regime to survive today, but it also makes any subsequent failure less informative – leaving participants more optimistic and willing to participate than they might have been otherwise. As a result, repression can be self-defeating. When mobilization is sensitive to beliefs, organizers are keen to preserve tactics that are robust to declines in confidence. For the regime, the same responsiveness can mean that repression is especially prone to backfiring: A weaker failure preserves optimism, substantially bolstering subsequent turnout and giving the movement more capacity to re-mobilize in the future.

The paper is organized as follows: Section 2 discusses related literature. Section 3 presents the baseline model. Section 4 establishes core results without tactical choice. Section 5 analyzes tactical choice. Section 6 presents the repression extension. Section 7 concludes.

2 Literature

The model contributes directly to the formal literature on protest and coordination under incomplete information. I build on the global games approach Morris and Shin (1998, 2003), which uses small perturbations to common information in order to discipline equilibrium multiplicity. A substantial literature has used this approach to study protest, regime change, and political coordination (Boix and Svolik 2013; Casper and Tyson 2014; Edmond 2013; Rundlett and Svolik 2016; Shadmehr and Bernhardt 2017; Tyson and Smith 2018). Using global games to formulate dynamic predictions is difficult, however, because accumulated private information interacts with the public information conveyed by the public signal of regime survival, restoring multiplicity (Angeletos, Hellwig and Pavan 2007; Little 2017; Shadmehr 2019). Existing approaches address this problem through devices such as forward-looking altruism (Gieczewski and Koçak 2024) or staggered entry (Dasgupta 2007). I take a different approach by assuming participation is driven by private signals of participation *costs*, rather than by public beliefs about the regime updated from observed failures. This preserves uniqueness while maintaining a transparent relationship between public beliefs and participation.

The model is also closely related to formal work on contentious politics with multiple tactics or repertoires (Boyer et al. 2026; Bueno de Mesquita 2013; Morris and Shadmehr 2024; Qiu 2022). Most closely, Bueno de Mesquita (2013) models tactics in essentially the same technological terms used here: they differ in how they translate participation into political power. The source of dynamics is substantially different, however. In Bueno de Mesquita (2013), organizational capacity is exogenous and tactical choice occurs after participation decisions. By contrast, in my model, organizational capacity is endogenous to public beliefs and tactical choice can affect the beliefs environment in which future tactics are chosen. Morris and Shadmehr (2024) also study repertoire choice, but focus on the one-shot decision to make a broader or narrower set of actions available rather than on the evolution of tactical choice across repeated attempts.

My research question is also superficially similar to Boyer et al. (2026), who study dynamic usage of peaceful and violent tactics, particularly within the context of the French “yellow vests” movement. However, in their model, peaceful or violent behavior reflect citizens’ underlying propensities for violence; dynamics arise from learning about the composition of the movement. In my model, tactics do not correspond to agent types, and learning concerns the strength of the regime. The same distinction also separates my paper from a broad empirical literature which studies how the usage of violent or radical tactics can change movement composition, participation costs, and outsiders’ perceptions (Chenoweth 2023; Chenoweth and Stephan 2011; Manekin and Mitts 2022; Simpson, Willer and Feinberg 2018; Steinert-Threlkeld, Chan and Joo 2022; Wasow 2020). I abstract away from audience or compositional effects, and instead study on what forms of action remain productive as mobilization itself rises and falls.

3 Model

Overview. There are two periods, $t \in \{1, 2\}$, and two sets of strategic actors: a protest organizer and a unit mass of potential participants. The organizer chooses the tactic used in each period in order to maximize the probability that the movement succeeds over its lifetime. The organizer is forward-looking and does not discount the future. The organizer cannot compel participation: after observing the chosen tactic, citizens make individually optimal participation decisions.

The movement confronts a regime of unknown strength θ . Before the game begins, θ is drawn from $\text{Unif}[0, M]$, with $M > 1$. Let F_t denote the commonly held beliefs at the beginning of period t , so $F_1 = \text{Unif}[0, M]$. Regime strength is fixed across periods. A successful protest ends the game; following a failed period 1 protest, the movement may try again in period 2.

Tactics and protest power. In each period in which the game remains active, the organizer chooses between two tactics: *peaceful* and *violent*, denoted by $\tau \in \{p, v\}$.¹ Each tactic is associated with a continuously differentiable and weakly increasing production function $A_\tau(\ell)$, which maps turnout $\ell \in [0, 1]$ – the share of potential participants who join the protest – into protest power. A protest using tactic τ succeeds if and only if $A_\tau(\ell) > \theta$.

For all $\ell \in [0, 1]$, assume:

(A1) $A_v(0) > A_p(0) = 0$: violence has higher baseline power;

(A2) $A_v(1) < A_p(1) = 1$: peace has higher full-participation power;

(A3) $A'_v(\ell) < A'_p(\ell)$: peaceful power is more responsive to additional participation.

A1 captures the idea that violent tactics can generate some power even with very little mass participation. A2 says that at the opposite end of full mobilization, peaceful tactics are more powerful. A3 ensures that this difference arises systematically as turnout increases. Together, the assumptions imply that the two technologies cross exactly once: there exists a unique $\ell^* \in (0, 1)$ such that $A_v(\ell) > A_p(\ell)$ for $\ell < \ell^*$, and vice versa for $\ell > \ell^*$.

Participation. Each potential participant i privately observes a period- t participation cost

$$c_{it} = c_t + \epsilon_{it},$$

where $c_t \sim \text{Unif}[-\epsilon, 1 + \epsilon]$ is a common component of participation costs and $\epsilon_{it} \sim \text{Unif}[-\epsilon, \epsilon]$ is idiosyncratic. The common shocks c_t are independent across periods. The idiosyncratic shocks

¹The restriction to two tactics rules out simultaneous deployment of different forms of contention by separate factions. The stylization captures environments in which movements coordinate around a predominant tactic in each period (Bueno de Mesquita 2013; Qiu 2022).

ϵ_{it} are i.i.d. across agents and periods.² Each citizen observes only their own realized cost c_{it} , and not the cost realizations of other citizens or the common shock.

The support of c_t guarantees two-sided dominance and ensures that conditioning on being the marginal participant does not truncate the support of the idiosyncratic shock. In particular, every possible participation cutoff lies at least ϵ away from the boundaries of the common-shock support. This delivers the standard Laplacian-beliefs property used below.

A citizen who participates receives benefit 1 if the protest succeeds and pays their participation cost regardless of the outcome. Thus,

$$u_{it} = \begin{cases} 1 - c_{it}, & \text{if } i \text{ participates and the protest succeeds,} \\ -c_{it}, & \text{if } i \text{ participates and the protest fails,} \\ 0, & \text{if } i \text{ does not participate.} \end{cases}$$

Timing. In each period in which the game remains active:

1. Given the public history and current beliefs F_t , the organizer chooses a tactic $\tau \in \{p, v\}$.
2. Participation costs are realized and privately observed. Citizens simultaneously decide whether to participate.
3. The protest occurs. Turnout ℓ_t and the protest outcome are publicly observed.
4. If $A_\tau(\ell_t) > \theta$, the protest succeeds and the game ends. If the protest fails in period 1, beliefs are updated and the game proceeds to period 2. The game ends after period 2 regardless of the outcome.

The solution concept is Perfect Bayesian Equilibrium.

4 Analysis without tactical choice

Before introducing tactical choice, I first establish the model's core linkage between beliefs, participation, and learning. Fix a single tactic function $A(\ell)$ and recall that a protest succeeds if and only if $A(\ell_t) > \theta$, with the game ending immediately upon success. After observing a failure, agents therefore learn that $\theta \geq A(\ell_t)$: all weaker regime types are ruled out. The common posterior is obtained by truncating the previous belief distribution from below at the realized power of the failed protest:

²Independence across periods abstracts from persistently committed "core activists" and from autocorrelated conditions such as sustained crackdowns. It ensures that period-2 participation responds to the history of the game through updated beliefs about regime strength rather than through persistence in the period-1 cost realization.

$$F_{t+1}(x) = \begin{cases} 0, & x < A(\ell_t), \\ \frac{F_t(x) - F_t(A(\ell_t))}{1 - F_t(A(\ell_t))}, & x \geq A(\ell_t) \end{cases} \quad (1)$$

My first result identifies how public beliefs translate into individual participation decisions.

Lemma 1 (Cutoff participation). *There exists a threshold $\hat{c}_t$ such that agent i participates in the period- t protest if and only if $c_{it} < \hat{c}_t$, where*

$$\hat{c}_t = \int_0^1 F_t(A(\ell)) d\ell \quad (2)$$

The cutoff is derived by considering the decision of an agent who is exactly indifferent between participating and not. The probability that the protest will succeed depends on turnout, which is induced by the common cost shock c_t , which the agent does not observe directly. Hence, she averages over the turnout levels that different values of c_t would induce:

$$\hat{c}_t = \int F_t(A(\ell(c_t))) g(c_t | c_i = \hat{c}_t) dc_t \quad (3)$$

I now apply the standard Laplacian-beliefs property of global games (Morris and Shin 2003). Conditional on being the marginal agent, $c_{it} = \hat{c}_t$, so $c_t = \hat{c}_t - \epsilon_{it}$. By the support assumption above, this conditioning does not truncate the support of ϵ_{it} . Conditional on being marginal, the agent is uncertain only about her rank in the population cost distribution; as the common shock varies uniformly over $(\hat{c}_t - \epsilon, \hat{c}_t + \epsilon)$, her rank – and hence induced turnout – ranges linearly from 0 to 1, yielding a uniform distribution of turnout over $[0, 1]$.³

Although citizens are forward-looking, their participation decisions are *as if myopic*, an observation made first in Angeletos, Hellwig and Pavan (2007) and Little (2017). Because each citizen is atomistic, her current participation decision has no effect on whether the regime survives, the beliefs inherited by the next period, or subsequent play. Her continuation payoff is therefore identical whether she participates or abstains, and cancels from the current participation decision.

Equation (2) shows that the marginal agent evaluates success by averaging uniformly over possible turnout levels. This then implies the distribution of actual turnout:

³The uniform turnout result relies on the uniform distribution of the common shock c_t but not on uniform idiosyncratic noise. More generally, for any continuous idiosyncratic distribution, the probability integral transform gives the same uniform rank belief. This is the standard Laplacian-beliefs property of global games (Morris and Shin 2003).

Lemma 2 (Equilibrium turnout). *Given participation cutoff $\hat{c}_t$, realized turnout is*

$$\ell_t = \begin{cases} 1 & \text{if } c_t \leq \hat{c}_t - \epsilon \\ \frac{\hat{c}_t - c_t + \epsilon}{2\epsilon} & \text{if } \hat{c}_t - \epsilon < c_t < \hat{c}_t + \epsilon \\ 0 & \text{if } c_t \geq \hat{c}_t + \epsilon \end{cases} \quad (4)$$

Since $c_t \sim \text{Unif}[-\epsilon, 1 + \epsilon]$, define

$$\rho := \frac{1}{1 + 2\epsilon}.$$

Then

$$\Pr(\ell_t = 1) = \rho \hat{c}_t, \quad \Pr(\ell_t = 0) = \rho(1 - \hat{c}_t),$$

while with remaining probability $1 - \rho$, turnout lies in $(0, 1)$ and is uniformly distributed conditional on being interior. Hence, $E[\ell_t \mid \hat{c}_t] = \frac{1-\rho}{2} + \rho \hat{c}_t$.

The Lemma has two noteworthy elements. First, ρ indexes how strongly realized turnout responds to changes in the participation cutoff: as ρ rises, more probability is concentrated at full or zero mobilization, and changes in beliefs have correspondingly larger effects on expected turnout. Secondly, it implies that a tactic's equilibrium probability of success is obtained by integrating over the realized turnout distribution. I characterize this object $S(A, F_t)$ explicitly in the Appendix.

The support assumptions ensure two-sided dominance and yield the Laplacian-beliefs property, which pins down a unique participation cutoff for any belief state (Lemma 1.) Lemma 2 then determines turnout for every realization of the common shock, while failure uniquely determines the posterior by truncation. Applying these arguments recursively across the two periods yields the unique equilibrium path:

Proposition 1 (Unique cutoff PBE). *There exists a unique Perfect Bayesian Equilibrium in threshold strategies. At every on-path history, agents participate in period t if and only if $c_{it} < \hat{c}_t$ with $\hat{c}_t$ given by (2). Beliefs update by truncation following failure. Given any realization of the common shocks $\{c_t\}_{t=1}^2$, these equilibrium strategies induce a unique path of turnout, protest outcomes, and posterior beliefs.*

These results make transparent the link between beliefs and mobilization. More pessimistic beliefs – F_t that place greater weight on strong regimes – shrink the participation cutoff. Tightening the cutoff then reduces expected turnout for the following period. The following corollaries formalizes these mechanisms:

Corollary 1 (Monotone decline of thresholds). *The cost threshold is weakly decreasing over time: $\hat{c}_{t+1} \leq \hat{c}_t$.*

Corollary 2 (Turnout declines in expectation). *Following a failed protest, $E[\ell_{t+1}] \leq E[\ell_t]$, with strict inequality whenever failure strictly lowers the participation cutoff.*

Note that *expected* decline does not preclude *realized* increases. A fortunate draw of the common shock can produce a turnout surge even after a failed first period. But on average, the movement is weaker tomorrow than today – not because participants have lost interest, but because they have learned that the task is harder than they thought.

Conditional on failure, larger protests are unambiguously more damaging to future beliefs than smaller protests, since they drive sharper pessimistic revisions. However, this does *not* mean that a movement hoping to maximize its chances at success would prefer low turnout to intermediate turnout. In the single-tactic setting, the direct benefit of large protests always compensates for the harsher informational consequences of failure:

Proposition 2 (Higher realized power improves lifetime success). *Fix a single tactic A that is used in both periods. The movement’s probability of succeeding by the end of period 2, conditional on realized period-1 power x , is weakly increasing and convex in x .*

The intuition comes from the informational cost of failure. Moving from a weak protest to a somewhat stronger one improves the movement’s chance of immediate success, but also makes a possible failure substantially more discouraging: more weak regime types are ruled out, leaving a worse continuation. Once the protest is already very strong, however, there is progressively less favorable continuation value left to destroy. Further increases in power continue to expand the set of regimes that can be defeated immediately, while imposing a smaller additional informational penalty if the protest still fails. Thus the direct payoff from greater power increasingly dominates its informational downside.

In the single-tactic setting, the movement has no control over this tradeoff: the available technology determines both the distribution of realized power and the belief states inherited after failure. Tactical choice introduces precisely such control, allowing the organizer to influence not only how much power the movement is likely to generate today, but also which post-failure belief states it is likely to face tomorrow. In the next section, I characterize precisely how organizers use this lever, and in particular how forward-looking concerns about failure shape tactical choice.

5 Tactical choice

Once the movement has access to more than one tactic, its organizer must decide not only which tactic is most effective under current conditions, but which tactic is most valuable to preserve for the mobilization environment that a failure may create. A tactic that performs especially well when turnout is high may be worth using immediately, while a tactic whose effectiveness survives demobilization may instead be most valuable in reserve. Tactical choice therefore depends on the movement’s comparative advantage across *future* mobilization states,

not simply on which tactic generates more power today.

I continue to study a success-maximizing organizer, who chooses the movement's tactic in each period to maximize the probability of eventual success. The organizer does not commit in advance to a tactical sequence, but rather makes an optimal choice in each period given the history up to that point. Three features determine how the organizer navigates the dynamic trade-off. Violence's baseline capacity determines how useful it is as insurance when mobilization is poor, while its responsiveness to turnout determines how much the movement gains from using it while mobilization is strong. Finally, the responsiveness of participation to beliefs determines how costly it is to lose favorable mobilization conditions after failure.

The organizer's problem somewhat resembles that of standard dynamic experimentation: a decision-maker chooses among actions or technologies not only for their immediate payoff, but also for what their outcomes reveal and how that information improves subsequent choices. In the typical setting, this creates a tradeoff between exploiting the action that currently appears best and experimenting in order to learn. Here, experimentation comes with an additional feedback loop: the results of those experiments also change efficacy with which technologies, or tactics will operate in the future. A failed protest produces more pessimistic beliefs about regime strength, which differentially reduces the relative productivity of tactics based on their dependence on turnout.

Before imposing additional structure on the tactical technologies, two observations help to sharpen intuitions about the sequencing problem. First, the asymmetry in tactical ceilings implies that some movements remain peaceful after every history, but no movements remain violent after every history. This asymmetry follows from the primitive structure of the technologies: after a sufficiently powerful failure, only the higher-ceiling peaceful technology can remain viable. This directly previews the continuation result of Proposition 3: sufficiently severe failures eventually favor peace, because they leave behind precisely the strong regime types against which violence's lower ceiling is most costly. But this does not by itself determine which tactic should be used first. In principle, the organizer could still open violently even if any failure would induce a subsequent switch to peace, because the opening tactic also determines the distribution of first-period successes and failures. The opening problem hinges on whether violence has enough of an advantage under favorable initial conditions to justify using it before the movement reaches the pessimistic states in which peace inevitably dominates.

The second observation identifies a more general sequencing principle. Consider a violent technology with arbitrarily high baseline power but no turnout sensitivity at all. The following lemma states that the organizer would never use such a tactic in the first period.

Lemma 3 (Flat violence is held in reserve). *If $A_v(\ell) \equiv \bar{v}$ for all ℓ , then a success-maximizing*

organizer never uses violence in period 1.

The logic is that a completely turnout-insensitive tactic loses nothing from being postponed. If violence is saved for period 2, its power remains $\bar{v}$ even after a failure has depressed public confidence and participation. Peaceful mass mobilization does not have this property: its effectiveness deteriorates when turnout falls. The organizer therefore uses the more “perishable” tactic first and preserves the tactic that is robust to demobilization for later. While stark, this benchmark isolates the central preservation motive without relying on the linear specification used below. A tactic can be more valuable when held in reserve than used immediately. Precisely because baseline capacity means that such a “flat” tactic is powerful *no matter* how weak or strong mobilization is, it is especially useful to save for the future. The linear case below introduces the competing force. Once violence itself becomes responsive to turnout, favorable mobilization conditions can also make violence especially attractive *now*.

5.1 Linear tactics

While the general model establishes that tactics can have intertemporal option value, because setbacks alter future mobilization, a linear setting facilitates decomposition of that option value into interpretable structural features, and characterization the resulting tactical paths. This also enables analysis of the wide range of cases in which both tactics are at least somewhat responsive to turnout.

Normalize the peaceful technology to $A_p(\ell) = \ell$, and let

$$A_v(\ell) = \alpha + \beta\ell, \quad \alpha > 0, \quad \beta \geq 0, \quad \alpha + \beta < 1.$$

The normalization is without loss of generality within the linear setting: the slope of peaceful power simply fixes the units in which protest power and regime strength are measured.

This specification separates two distinct sources of violent capacity. The intercept α captures the amount of power violence can generate even with essentially no mobilization, while the slope β captures how strongly violent power increases as turnout rises. The restriction $\alpha + \beta < 1$ preserves the assumption that peace has the higher ceiling under full mobilization. These forces then interact with a third parameter that governs the dynamics of mobilization. In effect, α provides a reason why violence may remain useful when mobilization is poor, whilst β provides a reason why violence may be especially valuable when mobilization is strong. The third parameter, ρ , determines how much the organizer values preserving favorable mobilization conditions for the future. These parameters provide a way of mapping observed features of movements and their environments into different incentives over tactical sequencing. They should not be interpreted only as reduced-form measures of whether violence is “strong,” but rather sources of variation which explain *when* violence is valuable.

The baseline capacity α captures the extent to which small groups can generate leverage with thin participation. Some forms of violent contention rely heavily on concentrated capabilities: a small handful of trained actors may be able to disable a rail junction or power installation, sabotage communications infrastructure, or exploit difficult terrain to impose costs on a much larger opponent. In such environments, numbers are only weakly related to what the tactic can accomplish. Dynamically, this makes violence especially valuable when mobilization has deteriorated, strengthening its role as a fallback after setbacks rather than necessarily creating a reason to use it immediately.

Violence’s turnout sensitivity β captures a different feature: the extent to which violent contention itself benefits from numbers. Street fighting or riots may become more effective when large crowds, potentially with access to arms, can overwhelm security cordons; and armed occupations or attempts to seize and hold urban territory may depend critically on assembling enough participants to control space. These are violent tactics whose effectiveness scales with mobilization. A high β therefore raises the return to using violence while participation is still strong, creating an incentive to deploy it early rather than preserve it solely as a fallback.

Finally, ρ captures how strongly changes in beliefs translate into changes in mobilization. Recall from Lemma 2 that, given participation cutoff $\hat{c}$, ρ measures how expected turnout varies with $\hat{c}$. When participation propensities are highly dispersed, mobilization is less sensitive to decreases in public confidence, corresponding to low ρ . In this case, changes in beliefs move relatively few citizens across the participation margin: turnout is comparatively resilient to pessimism. A layered movement drawing on a mix of committed activists, occasional participants, and marginal sympathizers fits this intuition: worsening prospects may discourage some participants without causing the entire mobilization base to disappear at once. Turnout is therefore comparatively resilient to pessimism.

When participation propensities are more tightly clustered, corresponding to high ρ , many citizens lie near the same participation margin. This might describe a broad but relatively homogeneous constituency whose members are willing to join when success appears plausible, but reluctant to participate once the movement begins to look doomed. A change in beliefs can then move a large share of the population into or out of participation at once. Setbacks therefore translate into sharp swings in turnout. Thus, high values of ρ makes those favorable conditions more perishable, strengthening the incentive to preserve a tactic that will remain effective after confidence and turnout have declined.

5.1.1 Responses to setbacks

I begin with the movement’s response to failure. Suppose an initial protest has failed: when does the organizer respond by turning to violence, and when does it instead continue with – or

return to – peaceful mass tactics? One might naturally expect the most serious setbacks to be the ones that induce escalation, since a powerful failed protest produces especially pessimistic beliefs and weakens subsequent participation. The model instead states that violence is used following sufficiently *weak* failures, not strong ones.

Proposition 3 (Continuation tactics). *There exists a cutoff*

$$c_\rho(\alpha, \beta; M) \in [0, \alpha + \beta)$$

such that violence is strictly preferred in period 2 following a failed first-period protest if and only if $x < c_\rho(\alpha, \beta; M)$. Whenever $c_\rho > 0$, the organizer is indifferent at $x = c_\rho$. The cutoff is weakly increasing in both α and β .

The result reflects a tension between *demobilization* and *selection on regime strength*. Stronger failures depress subsequent participation more, which by itself increases the appeal of a turnout-robust tactic. At the same time, however, they select increasingly strongly for regimes that can survive substantial protest power, against which violence’s lower ceiling is especially disadvantageous. In the linear setting, this second force dominates. Violence is consequently most attractive after a weak, disappointing failure: In these cases, the movement still has reason to believe that the regime may well be a weaker type. This is the state in which violence’s floor is valuable: it protects against another weak turnout realization while retaining a good chance of defeating the relatively weak regimes that remain possible. After a sufficiently powerful failure, the selection force dominates, and the organizer returns to the higher-ceiling peaceful technology.

The continuation response is not always path dependent. When the violent tactic is sufficiently weak, even a disappointing peaceful failure is not enough to make escalation worthwhile: peace remains optimal after every possible setback. As violent capacity improves, violence becomes useful as insurance in belief states where mobilization has gone badly – so some failed protests now induce escalation while stronger failures still lead back to peace. If violence has sufficiently high baseline capacity, this fallback role can arise even when violence itself is almost completely insensitive to turnout. Formally, for a given α , I denote by $\beta_0(\alpha)$ the point at which violence first becomes worthwhile after the weakest possible failure. Below this threshold the movement remains peaceful after every setback; above it, continuation tactical choice is path-dependent. The formal boundary is characterized algebraically in Appendix B.

The next question is which tactic the movement should use before any setback has occurred. Because tactics remain available in both periods, the organizer must choose an opening tactic knowing that failure changes the mobilization environment in which the next tactical choice is made. Holding baseline violent capacity α fixed, two forces pull in opposite directions. Greater scalability, β , increases the value of using violence in the first period: if violence itself benefits

strongly from high turnout, favorable initial mobilization is precisely when the organizer may want to use it. Greater mobilization responsiveness, ρ , instead strengthens the value of conditioning tactics on what the first protest reveals. Since mobilization is especially fragile, peaceful tactics are particularly appealing in the first period, as in the “flat violence” benchmark. Baseline capacity α affects both sides of this comparison. A higher α makes violence more powerful immediately, but it also increases the value of violence as insurance in low-mobilization states. Hence, it does not have an unambiguous effect toward using violence sooner or later.

I begin by comparing with the two limiting cases of mobilization responsiveness, which make the competing incentives to spend violence now or preserve it for later especially transparent. I then show that the same trade-off extends to intermediate values of ρ and characterize substantive implications.

5.1.2 Resilient mobilization: $\rho \rightarrow 0$

I begin with the limiting case in which changes in public confidence have little effect on realized mobilization. As idiosyncratic participation costs become increasingly dispersed, shifts in the participation cutoff move relatively few citizens across the margin. Turnout still varies substantially from protest to protest because of the common participation-cost shock, but that variation becomes essentially independent of the cutoff. For any fixed value of $\hat{c}$, turnout approaches $\text{Unif}[0, 1]$, and expected turnout approaches $1/2$ regardless of beliefs. In other words, although failures still change beliefs about regime strength and the participation cutoff, the link between those changing participation incentives and realized turnout is broken.

In this sense, this limiting case is closest to standard experimentation. The organizer continues to learn about an unknown objective – regime strength – but that learning no longer changes the distribution of mobilization under which the available tactical technologies will subsequently operate. As such, the preservation motive previewed by the flat violence benchmark is eliminated. Violence’s baseline capacity still makes it valuable as insurance against a poor turnout realization, but a failure today no longer makes such a realization systematically more likely tomorrow. So, there is less reason to preserve violence specifically for a post-setback state. The organizer therefore can place greater weight on the other dimension of the violent technology, β . When β is high, violence can capitalize on strong mobilization – holding it in reserve then means forgoing the opportunity to use a highly scalable tactic under favorable conditions.

The resulting equilibrium has three possible forms. When violence is simply a weak technology, the organizer uses peace throughout – there is no belief state in which the continuation is violent. As violent capacity improves, the organizer continues to open peacefully, but would use violence following sufficiently weak failures, as characterized by the continuation problem.

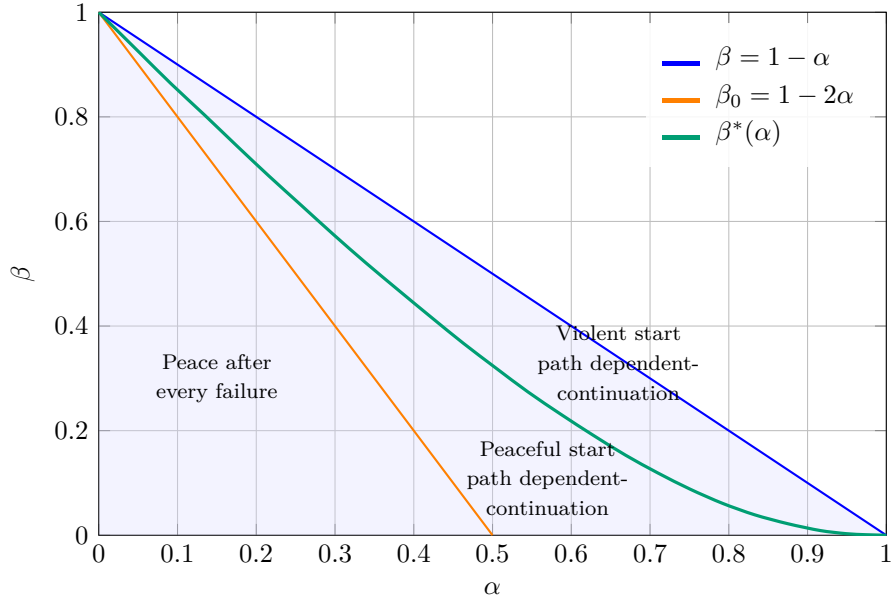


Figure 1. Tactical choice when $\rho \rightarrow 0$. The blue-shaded area is the feasible parameter region. Below the orange line, peace is optimal following every possible failure. Between the orange and dotted green lines, protests open peacefully, but the continuation is path dependent. Above the green line, violence is preferable in the first period, and the continuation is path dependent.

As violence becomes still more scalable, the balance eventually changes. The benefit of exploiting strong current mobilization outweighs the relatively small cost of giving up violence's fallback value, and the organizer becomes willing to use violence immediately. Thus, when $\rho \rightarrow 0$, there is a unique threshold $\beta^*(\alpha)$ such that violence is used as an opening tactic if $\beta > \beta^*(\alpha)$. The possible equilibrium paths can be summarized as follows:

	Opening tactic	Following failure
Weak violence	Peace	Peace
Intermediate violence	Peace	Violence after weak failure; peace after strong failure
Strong violence	Violence	Violence after weak failure; peace after strong failure

The missing fourth path is informative. The organizer never opens with violence only to abandon it after every possible failure. If violence is not robust enough to remain useful after even a weak failed violent protest, its value lies in reserve rather than as an opening gamble.

Figure 1 illustrates the progression. Moving rightward, greater baseline violent capacity makes violence useful in increasingly poor mobilization states, shrinking and eventually eliminating the region in which peace is used after every failure. Moving upward, greater turnout

sensitivity changes a different margin: violence becomes increasingly capable of exploiting strong mobilization itself. It first becomes valuable as an option following disappointing peaceful outcomes and, eventually, becomes valuable enough to deploy before any setback occurs.

The distinction between the comparative statics on α and β is important. Increasing α primarily improves violence precisely where peace is weakest – low-turnout states – and therefore strengthens violence’s role as a fallback. Increasing β instead improves violence in high-turnout states, where the movement must decide whether to preserve violence or take advantage of favorable conditions immediately. When mobilization is resilient to setbacks, there is relatively little cost to the latter choice, so violence is therefore eventually used immediately.

5.1.3 Fragile mobilization: $\rho \rightarrow 1$

Consider now the opposite limiting case, in which learning about the regime strongly shapes subsequent mobilization. When ρ is close to 1, potential participants respond very similarly to changes in beliefs. Rather than moving gradually in and out of participation, they tend to cross the participation margin together. Peaceful protest therefore becomes close to an all-or-nothing technology: realized turnout, and hence peaceful power, is usually either 0 or maximal.

This makes the feedback from learning to tactical effectiveness as strong as possible. A relatively minor deterioration in beliefs can generate a large decline in turnout, which in turn sharply reduces the power of tactics that depend heavily on mass participation. In this environment, a failed protest substantially changes the kind of tactical environment the organizer expects to face in the second period. Favorable mobilization conditions are therefore especially fragile, and the value of preserving tactics that remain effective after demobilization correspondingly high.

This limiting case also provides an endogenous reason why mass protest may appear to operate through a threshold-like technology. Classic threshold accounts of collective action build sharp changes in participation directly into individual behavior: once sufficiently many others participate, additional citizens cross their own participation thresholds and aggregate action can swing abruptly (Granovetter 1978; Kuran 1991). Here, by contrast, the peaceful production technology itself remains linear. The threshold-like pattern emerges from the mobilization process: when participation propensities are sufficiently clustered, relatively small changes in beliefs move large numbers of citizens across the participation margin together, so realized peaceful power becomes nearly all-or-nothing. Thus a technology that looks discontinuous at the aggregate level can arise even when neither the underlying production technology nor individual behavior contains such a discontinuity.

This threshold-like structure is also what gives peace its distinctive experimental value

in this limit: a poorly mobilized attempt fails while revealing almost nothing about regime strength, whereas a well-mobilized attempt delivers the strongest challenge the movement can mount. In effect, if a peaceful protest fails, it fails “cheaply” – with nearly no damage to beliefs. At the other extreme, a well-mobilized peaceful protest generates the strongest possible challenge to the regime. Violence has the opposite informational property. Because it has built-in baseline power, when a violent protest fails, beliefs become more pessimistic *even in states in which turnout itself was poor*. Furthermore, because ρ is high, that pessimism translates sharply into weaker future mobilization. These create strong incentives to use the turnout-dependent peaceful tactic while confidence is still high and to preserve the violent tactic that remains effective after demobilization – exactly the logic of the flat-violence benchmark.

Note the contrast with the low- ρ benchmark. A high β still generates a *technological* reason to use violence immediately: when violence scales strongly with turnout, favorable early mobilization conditions are precisely when it is most powerful. But a high ρ creates an opposing *dynamic* reason to preserve it. When mobilization is highly responsive to beliefs, a failed violent opening does not merely forgo success in the first period. Because violence generates substantial power even when turnout is poor, its failure produces bad news about regime strength which, in this limiting case, imposes a severe handicap on the peaceful tactic – and increases the value of the turnout-robust violent tactic that has already been “spent.” Peace, as discussed above, offers an unusually favorable first attempt in terms of both the upside (full turnout generates the strongest challenge available to the movement) and the downside (zero-turnout failures do little damage to beliefs.) As $\rho \rightarrow 1$, this asymmetry becomes sufficiently strong that the preservation motive dominates even when β is as high as possible.

Consider again the two boundaries illustrated in Figure 1 of previous $\rho \rightarrow 0$ benchmark. The β_0 boundary, describing the continuation role of violence, remains qualitatively unchanged – even when we take $\rho \rightarrow 1$, violence can be an appealing fallback technology if β is sufficiently high. By comparison, violence is less and less attractive as an opening tactic. Visually, imagine that as ρ approaches 1, this anticipatory preservation motive becomes strong enough that the green line is pushed up and out of the feasible range, thereby eliminating violent openings altogether. Consequently the first boundary survives while the second disappears, and the set possible equilibrium paths consequently collapses from three to two:

	Opening tactic	Following failure
Weak violence	Peace	Peace
Strong violence	Peace	Violence after weak failure; peace after strong failure

The following proposition describes how the violence-first region disappears between the

two limiting cases. For $\rho < \bar{\rho}$, sufficiently scalable violence can still justify an opening attack. As ρ rises, the required scalability β_ρ^* moves toward the feasibility boundary $\beta = 1 - \alpha$; at $\bar{\rho}$, even the most scalable admissible violent technology is no longer worth using first. Beyond that point, the continuation role of violence remains, but the violence-first region is absent.

Proposition 4 (Mobilization responsiveness and violent openings). *For every (α, M) , there exists a unique threshold $\bar{\rho}(\alpha, M) \in (0, 1)$ such that:*

- (i) *If $\rho < \bar{\rho}(\alpha, M)$, sufficiently scalable violence is used first. In particular, there exists a unique threshold $\beta_\rho^*(\alpha; M) < 1 - \alpha$ such that the organizer opens violently if and only if*

$$\beta > \beta_\rho^*(\alpha; M).$$

- (ii) *If $\rho \geq \bar{\rho}(\alpha, M)$, the organizer opens peacefully for every admissible violent technology.*

The preceding results suggest how structural differences across movements may map onto the equilibrium regions. The clearest examples of the exclusively peaceful region are movements whose coercive capacity derives overwhelmingly from visible mass participation. Some of the color revolutions – for example, the Georgian Rose Revolution of 2003 and the Ukrainian Orange Revolution of 2004 – fit this pattern well. Their leverage came from assembling very large numbers of ordinary citizens in central urban spaces and demonstrating that the incumbent could no longer govern in the face of sustained public opposition. When the violent tactic is too weak to function either as an appealing opening or reserve tactic, peace can therefore remain the movement’s preferred tactic throughout.

Even where violence is effective, it is only used as an opening tactic when two conditions coincide: Violence must be sufficiently scalable *and* movement’s mobilization must be relatively insulated from changes in public confidence. Only in these cases can violent openings can be used freely without sacrificing much future capacity. By contrast, if violence is scalable but mobilization is sensitive to confidence, it is preferable to use it as a reserve tactic only. Thus, the same coercive capacity can therefore generate different tactical sequences depending on the composition of potential participants.

6 Repression

I have so far treated the movement’s tactical environment as fixed. In practice, states actively shape that environment with repressive interventions, from closing public spaces to counterinsurgency measures. These do not only mechanically weaken protest, but also affect the sequencing incentives established above. I focus on two major implications: First, how organizers’ ability to substitute away from repressed tactics limits the direct efficacy of repression, and second, how repression changes the information conveyed by failure, potentially

making intervention partially self-defeating.

To establish these results, I introduce a regime actor whose objective opposes that of the protest organizer, retaining the linear tactical environment analyzed above. The regime loses 1 if the movement succeeds in either period and 0 if it survives both. In each period, it can weaken tactic τ by a factor $\gamma^\tau \in [0, 1]$, so that realized protest power becomes $x_t = \gamma_t^\tau A_\tau(\ell_t)$, where lower γ_t^τ denotes more intense repression. Repression incurs cost $\kappa_\tau(1 - \gamma_t^\tau)$, where κ_τ is continuous and strictly increasing, with $\kappa_\tau(0) = 0$. Finally, the regime incurs an additional cost $\lambda \geq 0$ in each period in which violence is used, capturing disruption, property damage, casualties, or other costs of violent contention independent of the immediate threat of overthrow.

Repression is publicly observed before participation decisions, so citizens account for it both when deciding whether to participate and when interpreting a subsequent failure. I assume further that the regime shares the movement’s uncertainty about its underlying strength, preserving the baseline truncation-based updating structure.⁴ I restrict the regime to targeting one tactic at a time, leaving the alternative technology unchanged; which tactic it targets, and how intensely, is part of its choice.

6.1 Responsive repression and tactical substitution

I begin in period 2, after an initial protest has failed. Because this is the terminal period, repression affects only the relative effectiveness of the two available tactics. The regime therefore has no reason to target the organizer’s fallback tactic while leaving its preferred tactic unchanged. As repression progressively weakens the preferred tactic, however, the alternative eventually becomes more attractive; once the organizer switches, further repression of the abandoned tactic has no additional effect on tactical choice or the movement’s probability of success.

Proposition 5 (Repression and tactical substitution). *Suppose that, following some first-period failure, the organizer strictly prefers one tactic while the alternative tactic remains viable. If the regime represses only the preferred tactic, there is a unique repression threshold at which the organizer switches to the alternative. Once that threshold is reached, further repression of the original tactic cannot further reduce the movement’s probability of success.*

The result caps the value of tactic-specific repression: repression can extract the preferred tactic’s advantage over its fallback, but no more. This interacts directly with the continuation result from the previous section. Absent repression, weak failures lead toward violence while stronger failures lead toward peace. Repressing violence following a weak failure can reverse this logic by inducing a switch toward peace. Repressing peace following a stronger failure

⁴Unobserved repression or private regime information would introduce additional inference and signaling problems that I abstract from here.

can analogously redirect the movement toward violence whenever violence remains a viable fallback. Thus repression and learning can generate *opposite patterns* of tactical adjustment.

The same logic gives a limited sense in which an unused violent option can protect peaceful protest (for instance, through violent flanks.) When peace is preferred but violence remains viable, crowd control cannot reduce the movement's continuation value below what the violent fallback can deliver. Latent violent capacity therefore places a floor under what repression of a peaceful movement can accomplish, even if it is never used.

6.2 Preemptive repression and the belief refund

I now turn to repression before the first protest, or *preemptive* repression. Such intervention does more than weaken the movement's immediate challenge: by reducing the power of a failed protest, it also changes what that failure reveals about regime strength. A weaker failure can therefore leave the movement more optimistic, and better able to mobilize, in the period that follows. To isolate this informational consequence, I hold period-2 repression fixed at zero. The previous subsection already characterizes how the regime can respond once a failure has occurred; this benchmark isolates preemptive repression changes the state from which that continuation begins. Because weaker failures leave movements more optimistic, repression consequently returns part of its immediate benefit to the movement through more favorable continuation beliefs:

Proposition 6 (The belief refund). *Suppose period-2 technologies are unrepressed. Lowering realized first-period protest power weakly improves the movement's continuation prospects. Moreover, consider sufficiently weak first-period challenges, $x \leq \min\{\alpha, c_\rho(\alpha, \beta; M)\}$. Within this region, lowering first-period power leaves the movement's eventual probability of success unchanged when $\rho\beta = 0$, but strictly increases eventual success when $\rho > 0$ and $\beta > 0$.*

The belief refund can do more than erase the regime's immediate gain from repression: when mobilization responds to beliefs, it can make repression *counterproductive*. The contrast is clearest in the two limiting cases. When $\rho = 0$, turnout is insensitive to beliefs, and repression can receive a *full refund* in the region of sufficiently weak first-period challenges. In this region, failure induces a switch to violence, whose baseline capacity is sufficient to defeat the marginal regime types spared by repression in period 1. When $\rho > 0$, however, repression can protect and even bolster mobilization. By weakening the first-period challenge, the regime makes a subsequent failure less discouraging. Because turnout responds to beliefs, the resulting optimism translates into higher second-period participation. Crucially, that additional participation can make the movement's fallback tactic more powerful. When $\beta > 0$, violence becomes more effective as more citizens participate. If turnout is high, the violent tactic can defeat the marginal regime types that repression spared in period 1, and even threaten still stronger regimes. The refund is then *more than complete*: reducing the movement's immediate power

can increase its eventual probability of success.

This is the regime-side counterpart of the preservation motive in the previous section. There, a forward-looking organizer values robust violence because failure destroys the mobilization conditions under which peaceful protest is effective. Preemptive repression partly prevents that destruction: by weakening a failure, the regime actually saves the optimism and mobilization that the movement would otherwise have lost.

The same informational logic has a direct implication for the persistence of violence. Consider a regime that represses an anticipated violent opening. As long as the organizer continues to open violently, stronger counterinsurgency makes the initial protest both more likely to fail, and to fail *weakly*. Weak failures generate precisely the states in which violence remains optimal in the continuation.

Proposition 7 (Weakening versus displacing violence). *Suppose repression affects only the period-1 violent technology and there is no period-2 repression. Among repression policies under which the organizer continues to open violently, stronger repression weakly increases the expected number of periods of violent protest. By contrast, any policy strong enough to induce a peaceful opening produces strictly less expected violence than any policy that preserves a violent opening.*

Repressing violence can therefore have sharply different effects depending on whether it merely weakens violence or changes the movement’s tactic. If the movement continues to begin violently, stronger repression shifts failures toward precisely the continuation states in which violence recurs. Once repression becomes strong enough to displace the violent opening, the certain first-period episode of violence disappears.

Taken together, the results identify two constraints on repression that mirror the two central forces in the baseline model. Movements can respond to tactic-specific repression by substituting toward an alternative technology, limiting how much the state can gain by targeting any single tactic. Furthermore, because failure shapes subsequent mobilization, temporary repression can preserve the beliefs and participation that a stronger failed protest would have destroyed.

7 Discussion

This paper develops a model of dynamic tactical choice in repeated protests. The central conceptual point is that tactics are differentially sensitive to their mobilization environment. At the same time, protest outcomes change beliefs about regime strength and therefore the participation that movements can sustain in the future. Tactical choice is consequently an intertemporal problem: movements choose among technologies whose relative value changes,

as contention itself creates new political conditions. This logic explains both why movements may switch tactics after failure, why the occurrence of these switches varies with the size of failures, and why organizers sometimes begin with tactics that are not expected to be particularly strong.

This perspective also changes how repression should be understood. Because organizers can substitute across tactics, repression of one technology may redirect contention toward another; because failed protests are informative, weakening a challenge also changes what its failure teaches citizens about the regime. These abilities of protests to adapt to and recover from repression constrain the value that regimes can extract; in some cases, they can even cause repression to backfire. More generally, the extension illuminated that movements *and* regimes shape the conditions under which subsequent contention will occur.

The same logic suggests implications beyond repression itself. One implication is that media freedom can sometimes be beneficial to regimes facing down protests. If failed protests demobilize partly through what they reveal about regime strength, accurate reporting of a visibly strong but unsuccessful challenge can be more discouraging than a coarse and obscured signal. A regime may sometimes value media freedom and transparency because it allows opposition setbacks to become common, and more discouraging knowledge. This observation is analogous to a large family of information design models concerning informational theories of repression.

A related implication concerns the value of commitment in repression. Greater policy flexibility may not always benefit the regime, because movements choose tactics in anticipation of how the state will respond later. If an organizer expects violence to be heavily repressed once it becomes an attractive fallback after failure, preserving violence for that future state becomes less valuable, potentially pushing the movement to use it immediately instead. A regime that could credibly commit to future forbearance might therefore induce a peaceful opening precisely by giving up an intervention it would prefer to use *ex post*. More generally, the model suggests that discretion can be costly when opponents reorganize their strategies around anticipated state responses. Characterizing the value of commitment to forbearance or rigidity is a natural direction for future work that builds upon the existing repression section.

Another direction concerns who chooses the movement's tactic. I have analyzed a unitary organizer, but there may instead be internal disagreement over who gets to make such strategic decisions. Once tactical choice is itself contested, the benchmark developed here provides a natural point of comparison. We might consider, for instance, a situation where multiple leaders bargain over strategy. Then, the sequence of tactics may depend on veto power, unilateral action, or the possibility of fragmentation. If, on the other hand, tactical choice is made in a decentralized way – for instance, through some kind of voting mechanism amongst all

participants – tactical choice may instead become sensitive to the evolving composition of the protest movement. I leave these extensions to future work.

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Appendix

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A Proofs for participation and benchmark results

A.1 Participation and turnout

Proof of Lemma 1. Fix beliefs F_t and a tactic with production function A . Because each citizen is atomistic, her current participation decision has no effect on whether the regime survives, the posterior inherited by the next period, or the continuation game. Continuation payoffs therefore cancel from the comparison between participating and abstaining, so participation is as-if myopic.

Let $\hat{c}_t$ denote the cost of a marginal citizen. Conditional on $c_{it} = \hat{c}_t$,

$$c_t = \hat{c}_t - \epsilon_{it}.$$

Since $\hat{c}_t \in [0, 1]$ and $c_t \sim \text{Unif}[-\varepsilon, 1 + \varepsilon]$, the interval $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$ lies inside the support of the common shock. Conditioning on marginality therefore does not truncate the support of ϵ_{it} . Moreover, because both c_t and ϵ_{it} have constant densities on their supports, the marginal citizen's posterior over c_t is uniform on $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$.

For any c_t in this interval, aggregate turnout under cutoff $\hat{c}_t$ is

$$\ell(c_t; \hat{c}_t) = \Pr(c_t + \epsilon_{jt} < \hat{c}_t) = \frac{\hat{c}_t - c_t + \varepsilon}{2\varepsilon}.$$

Thus, as the marginal citizen's posterior over c_t ranges uniformly over $[\hat{c}_t - \varepsilon, \hat{c}_t + \varepsilon]$, induced turnout ranges uniformly over $[0, 1]$. Her expected benefit from participating is therefore

$$\int_0^1 F_t(A(\ell)) d\ell.$$

Indifference of the marginal citizen gives

$$\hat{c}_t = \int_0^1 F_t(A(\ell)) d\ell,$$

which is equation (2). The right-hand side is determined by F_t and A , so the cutoff is unique. $\square$

Proof of Lemma 2. Fix a participation cutoff $\hat{c}_t \in [0, 1]$. Actual turnout is

$$\ell_t = \begin{cases} 1, & c_t \leq \hat{c}_t - \varepsilon, \\ \frac{\hat{c}_t - c_t + \varepsilon}{2\varepsilon}, & \hat{c}_t - \varepsilon < c_t < \hat{c}_t + \varepsilon, \\ 0, & c_t \geq \hat{c}_t + \varepsilon. \end{cases}$$

The support of c_t has length $1 + 2\varepsilon$. The set of common shocks generating full turnout has length $\hat{c}_t$, so

$$\Pr(\ell_t = 1) = \frac{\hat{c}_t}{1 + 2\varepsilon} = \rho\hat{c}_t.$$

The set generating zero turnout has length $1 - \hat{c}_t$, so

$$\Pr(\ell_t = 0) = \rho(1 - \hat{c}_t).$$

The remaining interval has length 2ε and hence probability $1 - \rho$. Conditional on this interior region, the mapping from c_t to ℓ_t is linear, so realized turnout is uniformly distributed on $(0, 1)$. Consequently,

$$\mathbb{E}[\ell_t \mid \hat{c}_t] = \rho\hat{c}_t + (1 - \rho)\frac{1}{2} = \frac{1 - \rho}{2} + \rho\hat{c}_t.$$

□

Lemma 2 implies that for a fixed tactic A , its probability of success under beliefs F_t is defined as follows:

Definition 1 (Actual success probability). *For beliefs F_t and tactic A , define the actual probability of success as*

$$S(A, F_t) = \rho(1 - \hat{c}_t)F_t(A(0)) + (1 - \rho) \int_0^1 F_t(A(\ell)) d\ell + \rho\hat{c}_t F_t(A(1)).$$

Since

$$\hat{c}_t = \int_0^1 F_t(A(\ell)) d\ell,$$

this can equivalently be written as

$$S(A, F_t) = \rho(1 - \hat{c}_t)F_t(A(0)) + (1 - \rho)\hat{c}_t + \rho\hat{c}_t F_t(A(1)).$$

Proof of Proposition 1. At every history, beliefs F_t and the tactic in use uniquely determine the citizens' participation cutoff through equation (2). Given this cutoff and a realization of the common shock c_t , Lemma 2 uniquely determines realized turnout and therefore realized protest power. Following a failed protest, Bayes' rule uniquely determines the truncated posterior.

Backward induction therefore uniquely determines the citizens' threshold strategies at every history. The organizer's tactic is also unique whenever one tactic is strictly preferred. At knife-edge histories at which the organizer is indifferent, the induced equilibrium is unique only up to the organizer's tie-breaking choice (or under any tie-breaking convention imposed in the main text). Away from such indifference, the equilibrium path of turnout, protest outcomes, and posterior beliefs is unique for every realization of the underlying state and common shocks. $\square$

Proof of Corollary 1. Suppose a period- t protest fails at realized power x . The posterior F_{t+1} is obtained by truncating F_t from below at x . Hence

$$F_{t+1}(y) \leq F_t(y) \quad \text{for every } y,$$

with strict inequality on a nonempty interval whenever the failure produces a nontrivial update. Equation (2) then implies

$$\hat{c}_{t+1} = \int_0^1 F_{t+1}(A(\ell)) d\ell \leq \int_0^1 F_t(A(\ell)) d\ell = \hat{c}_t.$$

The inequality is strict whenever the posterior shift affects a positive-measure set of power realizations generated by the tactic. $\square$

Proof of Corollary 2. By Lemma 2,

$$\mathbb{E}[\ell_t \mid \hat{c}_t] = \frac{1-\rho}{2} + \rho\hat{c}_t.$$

This expression is strictly increasing in $\hat{c}_t$. The result therefore follows immediately from Corollary 1. $\square$

A.2 Realized power and lifetime success

Proof of Proposition 2. Fix a single increasing tactic A that is used in both periods, and let $\bar{A} := A(1)$. Let $V(x)$ denote the movement's probability of succeeding by the end of period 2 conditional on realized period-1 power x .

Following failure at power x , the posterior is $\text{Unif}[x, M]$. Define

$$h(x) := \int_0^1 (A(\ell) - x)_+ d\ell, \quad G(x) := \int_0^1 \mathbf{1}\{A(\ell) > x\} d\ell.$$

The period-2 participation cutoff is

$$q(x) = \frac{h(x)}{M - x}.$$

The function h is convex and Lipschitz, hence absolutely continuous, with $h'(x) = -G(x)$ almost everywhere.

Because x itself was generated by tactic A , we have $x \geq A(0)$. Thus, following failure at x , zero turnout cannot overthrow any regime type remaining in the posterior. Full turnout succeeds with probability

$$r(x) := \frac{\bar{A} - x}{M - x}.$$

Using Lemma 2, the actual period-2 success probability is therefore

$$S_2(x) = (1 - \rho)q(x) + \rho q(x)r(x).$$

Under the prior $\text{Unif}[0, M]$, lifetime success conditional on period-1 power x is

$$V(x) = \frac{x}{M} + \frac{M - x}{M} S_2(x),$$

so

$$MV(x) = x + h(x)[1 - \rho + \rho r(x)]. \tag{A1}$$

Since

$$r'(x) = -\frac{1 - r(x)}{M - x},$$

differentiating (A1) at points of differentiability gives

$$MV'(x) = 1 - G(x) + \rho(1 - r(x))(G(x) - q(x)). \tag{A2}$$

Every term on the right-hand side is weakly nonnegative. In particular, $G(x) \leq 1$, $r(x) \leq 1$, and

$$q(x) = \int_{\{A(\ell) > x\}} \frac{A(\ell) - x}{M - x} d\ell \leq \int_{\{A(\ell) > x\}} 1 d\ell = G(x).$$

Hence $V'(x) \geq 0$ almost everywhere. Since V is absolutely continuous, V is weakly increasing.

For convexity, let

$$s(x) := 1 - r(x) = \frac{M - \bar{A}}{M - x}.$$

Then

$$s'(x) = \frac{s(x)}{M-x}, \quad q'(x) = -\frac{G(x) - q(x)}{M-x} \quad \text{a.e.}$$

The function G is weakly decreasing and therefore of bounded variation. Taking the distributional derivative of (A2) gives

$$M dV' = [1 - \rho s(x)](-dG) + \frac{2\rho s(x)}{M-x}(G(x) - q(x)) dx. \quad (\text{A3})$$

Both terms on the right-hand side are nonnegative measures: $-dG$ is nonnegative because G is decreasing; moreover, reachable power satisfies $x \leq \bar{A}$, so $s(x) \in [0, 1]$ and hence $1 - \rho s(x) \geq 0$; and $G(x) - q(x) \geq 0$. Thus the distributional second derivative of V is a nonnegative measure. Equivalently, V' is weakly increasing, and V is weakly convex over the full range of reachable first-period power. $\square$

A.3 Benchmark results with tactical choice

Proposition A1 (A sufficient condition for peace on the entire path). *Suppose peace is optimal following every possible period-1 failure. Let X_τ denote realized period-1 protest power when the organizer chooses tactic $\tau \in \{p, v\}$, under the equilibrium turnout distribution induced by that tactic.*

If, for every z ,

$$\mathbb{E}[(X_p - z)_+] \geq \mathbb{E}[(X_v - z)_+], \quad (\text{A1.1})$$

then peace is optimal in period 1 as well.

Proof of Proposition A1. Let $V_p(x)$ denote the movement's probability of succeeding by the end of period 2 conditional on realized period-1 power x and peaceful continuation following failure. By Proposition 2, V_p is weakly increasing and convex over the relevant range of realized power. The maintained floor and ceiling assumptions imply that the power range generated by violence lies inside this range for peaceful continuation.

Condition (A1.1) says that, above every threshold z , the peaceful opening has at least as much expected upper-tail power as the violent opening. This is the standard characterization of increasing-convex dominance of X_p over X_v . Equivalently, every increasing convex function g satisfies

$$\mathbb{E}[g(X_p)] \geq \mathbb{E}[g(X_v)].$$

Applying this to $g = V_p$ yields

$$\mathbb{E}[V_p(X_p)] \geq \mathbb{E}[V_p(X_v)].$$

Because continuation is peaceful after every failure, the two expectations are exactly the organizer's lifetime success probabilities from opening with peace and violence, respectively. Hence peace is optimal in period 1. $\square$

Proof of Lemma 3. Suppose $A_v(\ell) \equiv \bar{v}$ for all ℓ . Since $A_v(0) > A_p(0) = 0$, we have $\bar{v} > 0$.

Consider first an organizer who opens with violence. The first-period protest generates power $\bar{v}$ regardless of turnout. If it fails, every regime type remaining in the posterior satisfies $\theta \geq \bar{v}$, so flat violence cannot succeed in period 2. Conditional on opening with violence, optimal continuation is therefore peaceful.

Now compare this $v \rightarrow p$ sequence with the feasible strategy $p \rightarrow v$: open peacefully and, following failure, use flat violence in period 2. For every regime type $\theta < \bar{v}$, the $p \rightarrow v$ strategy sacrifices no probability of eventual success. Either peace succeeds in period 1 or, if it fails, flat violence in period 2 defeats the regime. Thus all types that can be defeated by flat violence are still defeated when the flat tactic is held in reserve.

For regime types $\theta \geq \bar{v}$, flat violence cannot succeed under either sequence. The movement's only opportunity to defeat these types therefore comes from its peaceful attempt. Under $p \rightarrow v$, that attempt occurs under the initial beliefs. Under $v \rightarrow p$, it occurs only after the regime has survived power $\bar{v}$. Because $\bar{v} > 0$, this failure produces a strictly more pessimistic posterior. By Corollary 1, the peaceful participation cutoff is therefore lower after failed violence; by Lemma 2, the resulting distribution of realized turnout is lower in the sense of first-order stochastic dominance. Since peaceful protest power is increasing in turnout, the peaceful attempt is strictly less effective when postponed.

It follows that

$$W_{p \rightarrow v} > W_{v \rightarrow p}.$$

But $v \rightarrow p$ is the optimal continuation strategy conditional on opening with violence, whereas after opening peacefully the organizer can do weakly better than the restricted $p \rightarrow v$ strategy by choosing its optimal period-2 tactic. Hence

$$W_1^p > W_1^v.$$

Therefore a success-maximizing organizer never uses flat violence in period 1. □

B Linear setting

The proof of the opening result proceeds in four steps. First, continuation choice is single-crossing in the power of the failed first-period protest. Second, a violent opening can be optimal only after violence becomes *repeatable*, meaning that it can remain optimal following a sufficiently weak failed violent opening. Third, once violence is repeatable, its opening advantage is strictly increasing in β . Finally, the sign of the opening comparison at the strongest feasible violent technology determines whether a violence-first region exists at all.

B.1 Step 1: continuation choice and the repeatability boundary

Following a failed first-period protest of realized power x , the posterior is $\text{Unif}[x, M]$. Let $S_p(x)$ and $S_v(x)$ denote the actual period-2 probabilities of success under peace and violence, respectively. It is useful to derive these objects explicitly, because the violent continuation problem changes form depending on whether the posterior lower bound x lies below or above violence's floor α .

Let F_x denote the posterior CDF and let $\hat{c}_2^\tau(x)$ denote the period-2 participation cutoff under tactic τ . From Lemma 1,

$$\hat{c}_2^\tau(x) = \int_0^1 F_x(A_\tau(\ell)) d\ell.$$

Given a cutoff $\hat{c}_2^\tau(x)$, Lemma 2 implies that the actual success probability can be written as

$$S_\tau(x) = \rho(1 - \hat{c}_2^\tau(x))F_x(A_\tau(0)) + (1 - \rho)\hat{c}_2^\tau(x) + \rho\hat{c}_2^\tau(x)F_x(A_\tau(1)).$$

The three terms correspond, respectively, to zero turnout, an interior turnout realization, and full turnout. The middle term simplifies because, conditional on interior turnout, ℓ is uniform on $(0, 1)$ and the average success probability is exactly the participation cutoff $\hat{c}_2^\tau(x)$.

For peace, $A_p(\ell) = \ell$. Since a failed first-period protest has $x \geq 0$,

$$\hat{c}_2^p(x) = \int_x^1 \frac{\ell - x}{M - x} d\ell = \frac{(1 - x)^2}{2(M - x)}, \quad F_x(A_p(0)) = 0, \quad F_x(A_p(1)) = \frac{1 - x}{M - x}.$$

Substitution therefore gives

$$S_p(x) = \frac{(1 - x)^2}{2(M - x)} \left[(1 - \rho) + \rho \frac{1 - x}{M - x} \right]. \quad (\text{A4})$$

For violence, first suppose $x \leq \alpha$. Even zero turnout then produces power at least as large as the posterior lower bound, so

$$\hat{c}_2^v(x) = \int_0^1 \frac{\alpha + \beta\ell - x}{M - x} d\ell = \frac{\alpha + \beta/2 - x}{M - x}.$$

Using

$$F_x(A_v(0)) = \frac{\alpha - x}{M - x}, \quad F_x(A_v(1)) = \frac{\alpha + \beta - x}{M - x},$$

and simplifying yields

$$S_v(x) = \frac{\alpha - x + (1 - \rho)\beta/2}{M - x} + \frac{\rho\beta(\alpha + \beta/2 - x)}{(M - x)^2}, \quad x \leq \alpha. \quad (\text{A5})$$

If instead $\alpha < x < \alpha + \beta$, low-turnout violent protests generate power below the posterior lower bound and therefore have no chance of success. Only turnout above $(x - \alpha)/\beta$ matters, so

$$\hat{c}_2^v(x) = \int_{(x-\alpha)/\beta}^1 \frac{\alpha + \beta\ell - x}{M - x} d\ell = \frac{(\alpha + \beta - x)^2}{2\beta(M - x)}.$$

Here $F_x(A_v(0)) = 0$ and $F_x(A_v(1)) = (\alpha + \beta - x)/(M - x)$, which gives

$$S_v(x) = \frac{(\alpha + \beta - x)^2}{2\beta(M - x)} \left[(1 - \rho) + \rho \frac{\alpha + \beta - x}{M - x} \right], \quad \alpha < x < \alpha + \beta. \quad (\text{A6})$$

Finally, if $x \geq \alpha + \beta$, even full-turnout violence cannot defeat any regime type remaining in the posterior, so $S_v(x) = 0$.

Lemma A1 (Single-crossing continuation choice). *For every $\beta > 0$, period-2 tactical choice is single-crossing in the realized power of the failed first-period protest. There is a cutoff $c_\rho(\alpha, \beta; M) \in [0, \alpha + \beta)$, with $c_\rho = 0$ denoting the case in which violence is never strictly preferred, such that*

$$S_v(x) > S_p(x) \iff x < c_\rho(\alpha, \beta; M).$$

At an interior cutoff, the organizer is indifferent. The cutoff is weakly increasing in both α and β .

Proof. The formula for violent success changes at $x = \alpha$, so the proof handles the two regions separately. When $x \geq \alpha$, violence's floor lies below the weakest regime type still considered possible, and the ratio $S_v(x)/S_p(x)$ has a convenient monotonic form. When $x \leq \alpha$, the violent floor itself can defeat some posterior types even at zero turnout; the same ratio is less transparent. In that region, we instead ask how large the baseline capacity α would have to be to make the organizer indifferent at each x , and show that this required value rises with x .

First consider $x \in [\alpha, \alpha + \beta)$. Dividing (A6) by (A4) gives

$$\frac{S_v(x)}{S_p(x)} = \frac{1}{\beta} \left(\frac{\alpha + \beta - x}{1 - x} \right)^2 \frac{(1 - \rho)(M - x) + \rho(\alpha + \beta - x)}{(1 - \rho)(M - x) + \rho(1 - x)}. \quad (\text{A7})$$

Because $\alpha + \beta < 1$, the first nonconstant ratio in (A7) is strictly decreasing in x . The second can be written as

$$\frac{C - x}{D - x}, \quad C = (1 - \rho)M + \rho(\alpha + \beta) < (1 - \rho)M + \rho = D,$$

and is also strictly decreasing. Hence $S_v(x)/S_p(x)$ is strictly decreasing on $[\alpha, \alpha + \beta)$. Since $S_v(\alpha + \beta) = 0 < S_p(\alpha + \beta)$, there is at most one crossing in this region, and any crossing is from violence to peace.

Now consider $x \leq \alpha$. Here the violent floor lies inside the posterior support, so even a zero-turnout violent protest can succeed against some of the remaining regime types. Rather than compare the two success functions by a ratio, fix (x, β, M, ρ) and solve for the baseline violent capacity that would make the organizer exactly indifferent. Because $S_v(x) - S_p(x)$ is affine and strictly increasing in α ,

$$\frac{\partial}{\partial \alpha}(S_v(x) - S_p(x)) = \frac{M + \rho\beta - x}{(M - x)^2} > 0.$$

There is therefore a unique indifference value $\alpha^*(x)$; this value need not itself be admissible. If $\alpha^*(x)$ is increasing in x , then for any fixed actual α the inequality $\alpha > \alpha^*(x)$ can switch from true to false at most once as x rises. This is exactly the desired single-crossing property in the low- x region. To establish that monotonicity, write

$$p := 1 - x, \quad k := M - 1, \quad u := \alpha^*(x) - x.$$

The indifference condition is equivalent to

$$2(k + p + \rho\beta)u = (1 - \rho)(k + p)(p^2 - \beta) + \rho(p^3 - \beta^2). \quad (\text{A8})$$

Implicit differentiation yields

$$\frac{d\alpha^*(x)}{dx} = \frac{P(p, \beta, k, \rho)}{2(k + p + \rho\beta)^2}, \quad (\text{A9})$$

where

$$\begin{aligned} P(p, \beta, k, \rho) &= \rho^2\beta(\beta + 2kp) \\ &\quad + \rho[\beta\{2k(2 - p) + p(4 - 3p)\} + kp(2k + p)] \\ &\quad + 2(k + p)^2(1 - p). \end{aligned} \quad (\text{A10})$$

Every term in (A10) is nonnegative, and the expression is strictly positive for admissible parameters. Thus $\alpha^*(x)$ is strictly increasing in x . Because violence is preferred exactly when $\alpha > \alpha^*(x)$, its relative advantage can cross zero at most once in the region $x \leq \alpha$, again from violence to peace.

The two expressions agree at $x = \alpha$, so the comparison is continuous there, although its derivative need not be. The two regions therefore combine into a single global cutoff.

It remains to establish the comparative statics of the continuation cutoff. Fix a posterior state x . It is enough to show that making the violent technology pointwise stronger cannot reduce its period-2 success probability at that posterior. Consider two violent technologies A_v and $\tilde{A}_v$ such that

$$\tilde{A}_v(\ell) \geq A_v(\ell) \quad \text{for every } \ell,$$

with strict inequality for some turnout levels. Raising α gives strict inequality for every ℓ , while raising β gives strict inequality for every $\ell > 0$. The participation cutoff under the stronger technology is therefore weakly higher:

$$\tilde{c}_2^v(x) = \int_0^1 F_x(\tilde{A}_v(\ell)) d\ell \geq \int_0^1 F_x(A_v(\ell)) d\ell = \hat{c}_2^v(x).$$

To translate this cutoff comparison into a comparison of actual success, use the same realization of the common participation-cost shock in the two environments. For any fixed common shock, the turnout formula in Lemma 2 is weakly increasing in the participation cutoff. Hence the stronger violent technology induces weakly higher realized turnout, $\tilde{\ell} \geq \ell$. Since $\tilde{A}_v$ is itself increasing in turnout and pointwise at least as strong as A_v ,

$$\tilde{A}_v(\tilde{\ell}) \geq \tilde{A}_v(\ell) \geq A_v(\ell).$$

Thus, holding fixed both the regime type and the common cost shock, every state in which the weaker violent technology succeeds is also a state in which the stronger one succeeds. Therefore $S_v(x)$ weakly increases when either α or β rises, while $S_p(x)$ is unchanged. The set of posterior states in which violence is optimal can only expand, so c_ρ is weakly increasing in both parameters. $\square$

Two continuation boundaries will be useful below. The first determines whether violence is ever optimal after failure. By single crossing, it is enough to compare the tactics following the weakest possible failure, $x = 0$. From (A4)–(A5), the sign of $S_v(0) - S_p(0)$ is the sign of

$$B_0(\beta) := \rho\beta^2 + [M(1 - \rho) + 2\rho\alpha]\beta + 2M\alpha - [M(1 - \rho) + \rho]. \quad (\text{A11})$$

Since

$$B'_0(\beta) = 2\rho\beta + M(1 - \rho) + 2\rho\alpha > 0,$$

this comparison is strictly increasing in β . Define

$$\alpha_0(M, \rho) := \frac{1 - \rho}{2} + \frac{\rho}{2M}.$$

At $\beta = 0$,

$$B_0(0) = 2M[\alpha - \alpha_0(M, \rho)].$$

Hence, if $\alpha < \alpha_0(M, \rho)$, violence loses to peace after the weakest possible failure when $\beta = 0$. As β rises, the comparison at $x = 0$ moves strictly in violence's favor. Define $\beta_0(\alpha; M, \rho)$ as the point at which violence first becomes just as good as peace after this weakest failure. Formally, it is the unique value in $(0, 1 - \alpha)$ satisfying $B_0(\beta_0) = 0$. Existence follows because $B_0(0) < 0$, whereas

$$B_0(1 - \alpha) = \alpha[M(1 + \rho) - \rho\alpha] > 0.$$

The single-crossing lemma then translates this one comparison at $x = 0$ into a statement about the entire continuation problem. For $\beta < \beta_0$, even the weakest failure is not enough to make violence attractive, so peace is strictly preferred following every possible failure. At $\beta = \beta_0$, the organizer is indifferent only following $x = 0$. For $\beta > \beta_0$, violence is optimal following sufficiently weak failures. Thus β_0 is the boundary at which violence first becomes useful as a *fallback* tactic. If $\alpha = \alpha_0(M, \rho)$, the boundary occurs at $\beta_0 = 0$. If $\alpha > \alpha_0(M, \rho)$, then $B_0(\beta) > 0$ for every admissible $\beta \geq 0$, so the exclusive-peace continuation region is absent.

The second boundary determines whether violence can remain optimal following a failed *violent* opening. Since the least powerful violent protest generates power α , this is exactly the point at which c_ρ crosses α . Let $\beta_\alpha(\alpha; M, \rho)$ denote the unique value satisfying $c_\rho = \alpha$. Substituting $x = \alpha$ into the continuation comparison gives

$$\beta [(1 - \rho)(M - \alpha) + \rho\beta] = (1 - \alpha)^2 [(1 - \rho)(M - \alpha) + \rho(1 - \alpha)]. \quad (\text{A12})$$

The left-hand side is strictly increasing in β and equals zero at $\beta = 0$. At $\beta = 1 - \alpha$ it strictly exceeds the right-hand side. Therefore

$$\beta_\alpha(\alpha; M, \rho) \in (0, 1 - \alpha)$$

is unique.

Whenever β_0 lies in the feasible parameter space, $\beta_0 < \beta_\alpha$. At $\beta = \beta_\alpha$, violence and peace are indifferent following $x = \alpha > 0$. Single crossing therefore implies that violence is strictly preferred following $x = 0$, so the value of β at which violence first becomes competitive at $x = 0$ must be strictly smaller.

Thus β_α is the relevant *repeatability boundary*. Below it, violence may be useful after a sufficiently weak peaceful failure, but cannot remain optimal after any failed violent opening. Above it, sufficiently weak failed violent openings can themselves be followed by violence.

B.2 Step 2: a violent opening requires repeatability

Under the initial prior, the period-1 participation cutoffs are

$$\hat{c}_1^p = \frac{1}{2M}, \quad \hat{c}_1^v = \frac{\alpha + \beta/2}{M}.$$

Let X_p and X_v denote realized period-1 protest power under peaceful and violent openings. Their CDFs on the interiors of their supports are

$$F_p(x) = \rho(1 - \hat{c}_1^p) + (1 - \rho)x, \quad 0 \leq x < 1, \quad (\text{A13})$$

and

$$F_v(x) = \begin{cases} 0, & x < \alpha, \\ \rho(1 - \hat{c}_1^v) + (1 - \rho)\frac{x - \alpha}{\beta}, & \alpha \leq x < \alpha + \beta, \\ 1, & x \geq \alpha + \beta. \end{cases} \quad (\text{A14})$$

The endpoint jumps correspond to the positive probabilities of zero and full turnout.

For proof purposes, define

$$H_p(x) := \frac{x}{M} + \frac{M - x}{M} S_p(x),$$

and define $H_v(x)$ analogously using $S_v(x)$. Thus $H_\tau(x)$ is lifetime success conditional on period-1 power x and continuation with tactic τ . Because $\alpha + \beta < 1$, the violent power support lies

inside $[0, 1]$, the relevant range for peaceful continuation. By Proposition 2, H_p is weakly increasing and convex throughout this range.

The repeatability boundary now has a direct implication for the opening decision. If $c_\rho \leq \alpha$, every possible violent opening produces power $X_v \geq \alpha \geq c_\rho$. Hence, conditional on failure, a violent opening is always followed by peace: violence can be used first, but it cannot be used again. A peaceful opening is different. After a sufficiently weak peaceful failure, the organizer may still switch to violence and use its positive floor in precisely the continuation states where it is most useful. The next lemma shows that, in this region, using violence first cannot improve lifetime success. Thus a violence-first equilibrium can arise only after violence becomes *repeatable* following some of its own weak failures.

Lemma A2 (A violent opening requires repeatability). *If*

$$c_\rho(\alpha, \beta; M) \leq \alpha,$$

then a success-maximizing organizer weakly prefers peace in period 1. The preference is strict whenever

$$0 < c_\rho(\alpha, \beta; M) \leq \alpha.$$

Equivalently, a strictly optimal violent opening requires $c_\rho(\alpha, \beta; M) > \alpha$.

Proof. Write $c := c_\rho(\alpha, \beta; M)$. If $\beta = 0$, the result follows from Lemma 3. Hence suppose $\beta > 0$.

Because every violent first-period realization satisfies $X_v \geq \alpha \geq c$, a failed violent opening is always followed by peace. A peaceful opening, by contrast, is followed by violence after failures with $X_p < c$ and by peace after failures with $X_p > c$. The proof compares these two opening payoffs directly.

First suppose $0 < c \leq \alpha$. In the region $x \leq \alpha$, (A5) implies

$$H_v(x) = \frac{\alpha + (1 - \rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - x)}{M(M - x)}, \quad (\text{A15})$$

so

$$H'_v(x) = \frac{\rho\beta(2\alpha + \beta - 2M)}{2M(M - x)^2} < 0, \quad (\text{A16})$$

where the inequality follows from $\alpha + \beta < 1 < M$.

Using Stieltjes integration by parts and the indifference condition $H_v(c) = H_p(c)$, the difference in opening payoffs can be written as

$$W_1^p - W_1^v = - \int_0^c F_p(x) H'_v(x) dx + \int_c^1 H'_p(x) [F_v(x) - F_p(x)] dx. \quad (\text{A17})$$

The first term is strictly positive by (A16). It therefore remains only to show that the second term is nonnegative.

From (A13)–(A14), $F_v - F_p$ is negative below α , strictly increasing on $[\alpha, \alpha + \beta)$, and positive above $\alpha + \beta$. Hence it crosses zero at most once, and any crossing is from negative to

positive. The second term in (A17) weights this difference by $H'_p(x)$. Because H'_p is nonnegative and increasing, the negative portion receives weakly less weight than the positive portion. It is therefore enough to show that, before applying these increasing weights, the positive area eventually outweighs the negative area over $[c, 1]$. The following claim establishes exactly that comparison.

Claim. If $0 < c \leq \alpha$ and $S_v(c) = S_p(c)$, then

$$\int_c^1 [F_v(x) - F_p(x)] dx > 0. \quad (\text{A18})$$

Because both opening-power distributions are supported on $[0, 1]$, the same inequality can be written as

$$\mathbb{E}[(X_p - c)_+] > \mathbb{E}[(X_v - c)_+].$$

Thus, measured relative to the continuation threshold c , a peaceful opening generates more expected power above that threshold. Equivalently, before the increasing weights $H'_p(x)$ are applied, the positive area where $F_v - F_p > 0$ outweighs the negative area where $F_v - F_p < 0$. Because the positive part occurs at higher values of x , weighting by the increasing function H'_p only strengthens this comparison.

Proof of the claim. Direct integration, followed by use of period-2 indifference at c , gives

$$\int_c^1 [F_v(x) - F_p(x)] dx = \frac{c}{2M} [1 + \rho + c^2 - \rho c - 2\alpha - (1 + \rho)\beta]. \quad (\text{A19})$$

It remains to sign the bracket. Let

$$z := 1 - c, \quad k := M - 1, \quad u := \alpha - c, \quad s := z - \beta.$$

Because $c \leq \alpha$ and $\alpha + \beta < 1$,

$$u \geq 0, \quad s - u = 1 - \alpha - \beta > 0.$$

The indifference condition can be written as

$$2(k + z + \rho\beta)u = U, \quad (\text{A20})$$

where

$$U = k(1 - \rho)[s - z(1 - z)] - z^2(1 - z) + zs(1 + \rho) - \rho s^2. \quad (\text{A21})$$

Since $u \geq 0$, $U \geq 0$. Substitution of (A20) into the bracket in (A19) yields

$$1 + \rho + c^2 - \rho c - 2\alpha - (1 + \rho)\beta = \frac{\rho}{k + z + \rho\beta} \{U + k[(1 + \rho)s - \rho z(1 - z)] + zs(1 - z)\}. \quad (\text{A22})$$

If $s \geq z(1 - z)$, every term in braces is nonnegative and the final term is strictly positive.

Suppose instead that $s < z(1 - z)$. The first term on the right-hand side of (A21) is then strictly negative. Since $U \geq 0$, it follows that

$$-z^2(1 - z) + zs(1 + \rho) - \rho s^2 > 0.$$

Writing $y := s/z \in [0, 1]$, this is

$$f(y) := -(1 - z) + (1 + \rho)y - \rho y^2 > 0.$$

On $[0, 1]$,

$$f'(y) = 1 + \rho - 2\rho y \geq 1 - \rho > 0,$$

and

$$f\left(\frac{\rho(1 - z)}{1 + \rho}\right) < 0.$$

Hence

$$(1 + \rho)s - \rho z(1 - z) > 0.$$

The terms in braces in (A22) are therefore again strictly positive. Since $\rho > 0$, the bracket in (A19) is strictly positive, proving (A18). $\square$

Return now to (A17). Let x_0 denote the (possibly degenerate) crossing point of $F_v - F_p$. Since H'_p is increasing, its weight on the positive part of $F_v - F_p$ is at least as large as its weight on the negative part. Together with (A18), this implies

$$\int_c^1 H'_p(x) [F_v(x) - F_p(x)] dx \geq 0.$$

The first term in (A17) is strictly positive, so

$$W_1^p > W_1^v$$

whenever $0 < c \leq \alpha$.

Finally, suppose $c = 0$. Then peace is weakly optimal following every possible failure. Moreover,

$$\int_0^1 [F_v(x) - F_p(x)] dx = \mathbb{E}[X_p] - \mathbb{E}[X_v] = M[S_p(0) - S_v(0)] \geq 0.$$

Because $F_v - F_p$ crosses at most once from negative to positive, this total integral inequality implies

$$\int_z^1 [F_v(x) - F_p(x)] dx \geq 0 \quad \text{for every } z.$$

In words, from every threshold z onward, the peaceful opening has at least as much upper-tail power on average as the violent opening. This one-crossing property is precisely what is needed for every increasing convex function of power to have weakly higher expectation under X_p . Since H_p is increasing and convex,

$$\mathbb{E}[H_p(X_p)] \geq \mathbb{E}[H_p(X_v)].$$

Continuation is peaceful after every failure when $c = 0$, so these are exactly the opening payoffs. Hence $W_1^p \geq W_1^v$. $\square$

B.3 Step 3: monotonicity once violence is repeatable

Define the opening advantage of violence by

$$\Delta_\rho(\alpha, \beta; M) := W_1^v - W_1^p.$$

The preceding lemma shows that a violence-first region, if it exists, must lie strictly above β_α . This substantially narrows what remains to be proved. We do *not* need the opening advantage of violence to rise with β everywhere. Below β_α , the previous lemma already tells us that peace opens. We only need the opening advantage Δ_ρ to be monotone in β after violence is repeatable, because this is the only region in which a violence-first equilibrium can occur. The next lemma establishes exactly that local monotonicity of the opening advantage.

Lemma A3 (Monotonicity once violence is repeatable). *If*

$$c_\rho(\alpha, \beta; M) > \alpha,$$

then

$$\frac{\partial}{\partial \beta} \Delta_\rho(\alpha, \beta; M) > 0.$$

Proof. Write

$$c := c_\rho(\alpha, \beta; M), \quad \sigma := \frac{c - \alpha}{\beta} \in (0, 1).$$

The variable σ is simply the turnout level at which a violent protest generates exactly the continuation cutoff: $c = \alpha + \beta\sigma$. Thus, for a violent opening, turnout below σ generates a sufficiently weak failure that violence is used again, while turnout above σ leads the organizer to switch to peace. This change in continuation behavior is why the payoff derivative is somewhat more involved than the underlying intuition.

For later use, record the lifetime-success functions explicitly. The superscripts L and H distinguish whether the failed protest power lies below or above the violent floor α :

$$H_p(x) = \frac{x}{M} + \frac{1-\rho}{2M}(1-x)^2 + \frac{\rho}{2M} \frac{(1-x)^3}{M-x}, \quad (\text{A23})$$

$$H_v^L(x) = \frac{\alpha + (1-\rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - x)}{M(M-x)}, \quad x \leq \alpha, \quad (\text{A24})$$

and

$$H_v^H(x) = \frac{x}{M} + \frac{1-\rho}{2\beta M}(\alpha + \beta - x)^2 + \frac{\rho}{2\beta M} \frac{(\alpha + \beta - x)^3}{M-x}, \quad \alpha < x < \alpha + \beta. \quad (\text{A25})$$

Because $c > \alpha$, both openings can generate failures after which violence is used again. The following payoff expressions simply partition the period-1 power distribution according to which continuation tactic is selected. The terms multiplied by ρ are the endpoint masses at

zero or full turnout; the integrals multiplied by $1 - \rho$ average over the uniform interior-turnout region:

$$W_1^p = \rho(1 - \hat{c}_1^p)H_v^L(0) + (1 - \rho) \left[\int_0^\alpha H_v^L(x) dx + \int_\alpha^c H_v^H(x) dx + \int_c^1 H_p(x) dx \right] + \rho \hat{c}_1^p H_p(1), \quad (\text{A26})$$

$$W_1^v = \rho(1 - \hat{c}_1^v)H_v^L(\alpha) + \frac{1 - \rho}{\beta} \left[\int_\alpha^c H_v^H(x) dx + \int_c^{\alpha+\beta} H_p(x) dx \right] + \rho \hat{c}_1^v H_p(\alpha + \beta). \quad (\text{A27})$$

Define the fixed- x derivatives

$$K_L(x) := \frac{\partial H_v^L(x)}{\partial \beta} = \frac{1 - \rho}{2M} + \frac{\rho(\alpha + \beta - x)}{M(M - x)}, \quad (\text{A28})$$

$$K_H(x) := \frac{\partial H_v^H(x)}{\partial \beta} = \frac{(1 - \rho)y(2\beta - y)}{2M\beta^2} + \frac{\rho y^2(3\beta - y)}{2M\beta^2(M - x)}, \quad y := \alpha + \beta - x. \quad (\text{A29})$$

For a violent opening with interior turnout realization ℓ , let

$$J_\rho(\ell) := \frac{d}{d\beta} H_v^H(\alpha + \beta\ell; \beta) = K_H(\alpha + \beta\ell) + \ell \partial_x H_v^H(\alpha + \beta\ell; \beta).$$

Differentiate (A26)–(A27). Two things change when β rises: violence becomes directly more effective, and the continuation cutoff c moves. The second effect creates boundary terms when the integrals are differentiated. Those terms cancel, however, because at the moving boundary the organizer is indifferent between the two continuation tactics,

$$H_v^H(c; \beta) = H_p(c).$$

This envelope-style cancellation is useful: after it, we only need to sign the direct effect of β on the opening comparison. Changing variables $x = \alpha + \beta\ell$ in the relevant interior integrals gives

$$\frac{\partial \Delta_\rho}{\partial \beta} = (1 - \rho)\mathcal{C}_\rho(\sigma) + \rho\mathcal{E}_\rho, \quad (\text{A30})$$

The decomposition mirrors the turnout distribution from Lemma 2. The term $\mathcal{C}_\rho$ collects the effect of raising β through the continuous interior part of the turnout distribution, which occurs with probability $1 - \rho$. The term $\mathcal{E}_\rho$ collects the corresponding effect through the endpoint masses at zero and full turnout, whose total probability is ρ . The objects themselves still depend on ρ because continuation values and participation cutoffs depend on mobilization responsiveness; the factors $1 - \rho$ and ρ outside them simply reflect how much probability the turnout distribution places on the interior and endpoint components. Thus (A30) can be read as

change through interior turnout realizations + change through endpoint turnout realizations.

More explicitly,

$$\begin{aligned}\mathcal{C}_\rho(\sigma) &= \int_0^\sigma [J_\rho(\ell) - \beta K_H(\alpha + \beta\ell)] d\ell \\ &\quad + \int_\sigma^1 \ell H'_p(\alpha + \beta\ell) d\ell - \int_0^\alpha K_L(x) dx,\end{aligned}\tag{A31}$$

and

$$\begin{aligned}\mathcal{E}_\rho &= \frac{H_p(\alpha + \beta) - H_v^L(\alpha)}{2M} + (1 - \hat{c}_1^v)K_L(\alpha) + \hat{c}_1^v H'_p(\alpha + \beta) \\ &\quad - (1 - \hat{c}_1^p)K_L(0).\end{aligned}\tag{A32}$$

Both $\mathcal{C}_\rho$ and $\mathcal{E}_\rho$ are affine in ρ . Write

$$\mathcal{C}_\rho = (1 - \rho)\mathcal{C}_0 + \rho\mathcal{C}_1, \quad \mathcal{E}_\rho = (1 - \rho)\mathcal{E}_0 + \rho\mathcal{E}_1.$$

Then

$$\frac{\partial \Delta_\rho}{\partial \beta} = (1 - \rho)^2 \mathcal{C}_0(\sigma) + \rho(1 - \rho) [\mathcal{C}_1(\sigma) + \mathcal{E}_0] + \rho^2 \mathcal{E}_1.\tag{A33}$$

Equation (A33) separates the derivative into three pieces with nonnegative weights $(1 - \rho)^2$, $\rho(1 - \rho)$, and ρ^2 . It is therefore enough to show that each associated coefficient is positive. Nothing conceptual changes in the remaining algebra, so we isolate the three sign checks as claims.

Claim 1: $\mathcal{C}_0(\sigma) > 0$. This is the coefficient associated with the $\rho = 0$ benchmark, in which all turnout realizations are interior. Direct integration of (A31) gives

$$\mathcal{C}_0(\sigma) = \frac{2\beta + 3(1 - \beta)\sigma - 3\alpha\sigma^2 + (1 - \beta)\sigma^3}{6M}.\tag{A34}$$

Since $\alpha < 1 - \beta$,

$$6M\mathcal{C}_0(\sigma) > 2\beta + (1 - \beta)(3\sigma - 3\sigma^2 + \sigma^3) = 2\beta + (1 - \beta)[1 - (1 - \sigma)^3] > 0.$$

Claim 2: $\mathcal{C}_1(\sigma) + \mathcal{E}_0 > 0$. This is the mixed coefficient, combining the effect of ρ on the interior part of the turnout distribution with the first endpoint contribution. We first show that $\mathcal{C}_1$ is strictly increasing in σ , which lets us bound it by its value at $\sigma = 0$. Let

$$e := 1 - \alpha - \beta > 0, \quad k := M - 1 > 0, \quad u := 1 - \sigma \in (0, 1).$$

Differentiating (A31) at $\rho = 1$ gives

$$\mathcal{C}'_1(\sigma) = \frac{kA(u) + B(u)}{2M(k + e + \beta u)^2},\tag{A35}$$

where

$$\begin{aligned} A(u) &= 2\beta(1-\beta)u^3 + 6\beta e u(1-u) + 3e^2(1-u), \\ B(u) &= \beta^2(1-\beta)(u^4 + u^3) + \beta e u^2 [3\beta + u(2-5\beta)] \\ &\quad + 6\beta e^2 u(1-u) + 2e^3(1-u). \end{aligned}$$

All displayed terms are nonnegative. The only bracket whose sign is not immediate is $3\beta + u(2-5\beta)$; it is linear in u and is positive at both endpoints,

$$3\beta > 0, \quad 3\beta + (2-5\beta) = 2(1-\beta) > 0.$$

Hence $\mathcal{C}'_1(\sigma) > 0$.

At $\rho = 1$,

$$H'_p(x) = \frac{x}{M} + \frac{(M-1)(1-x)(2M-x-1)}{2M(M-x)^2} > \frac{x}{M},$$

and

$$K_L(x) = \frac{\alpha + \beta - x}{M(M-x)} \leq \frac{\alpha + \beta}{M^2}.$$

Therefore

$$\mathcal{C}_1(\sigma) > \mathcal{C}_1(0) > \frac{\alpha}{2M} + \frac{\beta}{3M} - \frac{\alpha(\alpha + \beta)}{M^2}. \quad (\text{A36})$$

At $\rho = 0$, direct calculation of (A32) gives

$$\mathcal{E}_0 = \frac{5\alpha^2 + 8\alpha\beta - 4\alpha + 3\beta^2 - 2\beta + 2}{4M^2}. \quad (\text{A37})$$

Combining (A36) and (A37),

$$\mathcal{C}_1(\sigma) + \mathcal{E}_0 > \frac{6M\alpha + 4M\beta + 3\alpha^2 + 12\alpha\beta - 12\alpha + 9\beta^2 - 6\beta + 6}{12M^2}. \quad (\text{A38})$$

The numerator is increasing in M . Replacing M by 1 and substituting $e = 1 - \alpha - \beta$ gives the lower bound

$$3 + 10\beta + 3e^2 - 6\beta e = 3 + 3e^2 + \beta(10 - 6e) > 0,$$

because $e \in (0, 1)$. Thus $\mathcal{C}_1(\sigma) + \mathcal{E}_0 > 0$.

Claim 3: $\mathcal{E}_1 > 0$. This is the remaining endpoint coefficient, corresponding to the $\rho = 1$ limit. The expression is less transparent directly, so we reparameterize using the slack in the violent ceiling and the distance between that ceiling and the top of the prior. Let

$$d := 1 - \alpha - \beta > 0, \quad D := M - \alpha - \beta.$$

Since $M > 1$, $D > d$. Direct simplification of (A32) at $\rho = 1$ gives

$$M^2 \mathcal{E}_1 = T(\beta, d, D),$$

where

$$T(\beta, d, D) = \frac{d^3}{4D} + \frac{\beta(D + \beta/4)}{D + \beta} - \frac{(1 - d - \beta/2)d^2(3D - d)}{2D^2} + \frac{1 - d}{2(D + 1 - d)}. \quad (\text{A39})$$

Treating (β, d, D) as coordinates and holding (d, D) fixed,

$$\frac{\partial T}{\partial \beta} = \frac{D^2 + D\beta/2 + \beta^2/4}{(D + \beta)^2} + \frac{d^2(3D - d)}{4D^2} > 0. \quad (\text{A40})$$

Hence $T(\beta, d, D) > T(0, d, D)$. Let $\eta := d/D \in (0, 1)$. At $\beta = 0$,

$$T(0, d, D) = \frac{\eta}{4[d + \eta(1 - d)]} N(d, \eta), \quad (\text{A41})$$

where

$$N(d, \eta) = 2d(1 - d)^2\eta^2 + 3d(1 - d)(3d - 2)\eta + 7d^3 - 6d^2 - 2d + 2. \quad (\text{A42})$$

The quadratic in η has vertex

$$\eta^*(d) = \frac{3(2 - 3d)}{4(1 - d)}.$$

If $d \leq 2/5$, then $\eta^*(d) \geq 1$, so the minimum over $\eta \in [0, 1]$ occurs at $\eta = 1$, where

$$N(d, 1) = 5d^2 - 6d + 2 > 0.$$

If $2/5 < d < 2/3$, the vertex is interior and

$$N(d, \eta^*) = -\frac{(5d - 4)(5d^2 - 8d + 4)}{8} > 0,$$

since $5d - 4 < 0$ and $5d^2 - 8d + 4 > 0$. If $d \geq 2/3$, then $\eta^*(d) \leq 0$, so the minimum occurs at $\eta = 0$. In this case

$$N(d, 0) = g(d) := 7d^3 - 6d^2 - 2d + 2.$$

The only critical point of g in $[2/3, 1]$ is

$$d^* = \frac{6 + \sqrt{78}}{21},$$

and

$$g(d^*) = \frac{486 - 52\sqrt{78}}{441} > 0.$$

Thus $N(d, \eta) > 0$ throughout $(d, \eta) \in (0, 1)^2$, so $T(0, d, D) > 0$. Equation (A40) then implies $\mathcal{E}_1 > 0$.

All three coefficients in (A33) are strictly positive. Therefore

$$\frac{\partial \Delta_\rho}{\partial \beta} > 0$$

whenever $c_\rho > \alpha$. □

B.4 Step 4: existence of a violence-first region

The first three steps reduce the global opening problem to a simple endpoint test. At $\beta = \beta_\alpha$, peace is strictly better as an opening tactic. Above that boundary, the opening advantage of violence rises strictly with β . Therefore a violence-first region exists if and only if violence has become better by the time β reaches its largest feasible value, $1 - \alpha$. Mobilization responsiveness ρ matters because it determines the sign of exactly this strongest-violence comparison.

Proof of Proposition 4. By Lemma A2,

$$\Delta_\rho(\alpha, \beta_\alpha; M) < 0.$$

By Lemma A3, Δ_ρ is strictly increasing in β throughout

$$(\beta_\alpha, 1 - \alpha).$$

Thus there can be at most one opening cutoff in the region in which a violent opening can occur. Whether such a cutoff exists is determined entirely by the opening comparison at the strongest feasible violent technology.

Define

$$\Phi(\alpha, M, \rho) := \lim_{\beta \uparrow 1 - \alpha} \Delta_\rho(\alpha, \beta; M).$$

At the boundary $\beta = 1 - \alpha$,

$$A_v(\ell) = \alpha + (1 - \alpha)\ell \geq \ell = A_p(\ell),$$

with strict inequality for $\ell < 1$. Thus violence weakly dominates peace pointwise in the continuation problem and is optimal in period 2 following every failure below power 1.

At this limiting technology, the participation cutoffs do not depend on ρ . Conditional lifetime-success values are affine in ρ , and the period-1 turnout distribution is also affine in ρ . Hence $\Phi(\alpha, M, \rho)$ is a polynomial of degree at most two in ρ . Direct evaluation at the two limiting mobilization environments gives

$$\Phi(\alpha, M, 0) = \frac{\alpha(1 - \alpha)}{6M} > 0 \tag{A43}$$

and

$$\Phi(\alpha, M, 1) = -\frac{\alpha(1 - \alpha)(M - 1)(2M - \alpha - 1)}{4M^3(M - \alpha)} < 0. \tag{A44}$$

Therefore Φ has at least one root in $(0, 1)$. Because it is a polynomial of degree at most two and has opposite signs at 0 and 1, it can have only one root in $(0, 1)$. Denote this unique root by

$$\bar{\rho}(\alpha, M) \in (0, 1).$$

Suppose first that

$$\rho < \bar{\rho}(\alpha, M).$$

Then $\Phi(\alpha, M, \rho) > 0$. Since

$$\Delta_\rho(\alpha, \beta_\alpha; M) < 0$$

and Δ_ρ is continuous and strictly increasing in β throughout $(\beta_\alpha, 1 - \alpha)$, there exists a unique

$$\beta_\rho^*(\alpha; M) \in (\beta_\alpha(\alpha; M, \rho), 1 - \alpha)$$

such that

$$\Delta_\rho(\alpha, \beta_\rho^*; M) = 0.$$

Hence

$$W_1^v > W_1^p \iff \beta > \beta_\rho^*(\alpha; M).$$

For $\beta \leq \beta_\alpha$, Lemma A2 already implies that peace is weakly preferred in period 1. Thus this is the unique violence-first region over the full feasible parameter space, establishing part (i) of Proposition 4.

Now suppose

$$\rho > \bar{\rho}(\alpha, M).$$

Then $\Phi(\alpha, M, \rho) < 0$. Since the opening advantage is negative at β_α and strictly increasing thereafter, it remains negative throughout

$$(\beta_\alpha, 1 - \alpha).$$

Below β_α , Lemma A2 implies that peace is weakly preferred. Hence peace is optimal in period 1 for every admissible $\beta < 1 - \alpha$.

The same conclusion holds at $\rho = \bar{\rho}(\alpha, M)$. At this value,

$$\lim_{\beta \uparrow 1 - \alpha} \Delta_{\bar{\rho}}(\alpha, \beta; M) = 0,$$

while strict monotonicity above β_α implies

$$\Delta_{\bar{\rho}}(\alpha, \beta; M) < 0$$

for every admissible interior $\beta < 1 - \alpha$. Again, Lemma A2 covers the remaining region. This establishes part (ii) of Proposition 4.

The argument also yields two useful additional characterizations. First, whenever a violence-first region exists,

$$\beta_\alpha < \beta_\rho^* < 1 - \alpha.$$

Whenever the exclusive-peace continuation boundary β_0 lies in the feasible parameter space, the continuation analysis above gives

$$\beta_0 < \beta_\alpha < \beta_\rho^* < 1 - \alpha.$$

Second, the violence-first threshold converges to the feasibility boundary as the critical level of mobilization responsiveness is approached:

$$\lim_{\rho \uparrow \bar{\rho}(\alpha, M)} \beta_{\rho}^*(\alpha; M) = 1 - \alpha.$$

To see this, let $\rho_n \uparrow \bar{\rho}(\alpha, M)$ and write

$$\beta_n := \beta_{\rho_n}^*(\alpha; M).$$

Any convergent subsequence of $\{\beta_n\}$ has a limit in

$$[\beta_{\alpha}(\alpha; M, \bar{\rho}), 1 - \alpha].$$

Continuity of β_{α} follows from equation (A12), while continuity of $\Delta_{\rho}(\alpha, \beta; M)$ follows from the continuous payoff formulas. An interior limit would therefore imply an interior zero of

$$\Delta_{\bar{\rho}}(\alpha, \beta; M),$$

which the preceding argument rules out. Hence every convergent subsequence has limit $1 - \alpha$, proving the claim. $\square$

C Proofs for repression extension

This section supplies proofs and supplementary results for the repression extension. Throughout, after a first-period failure at realized power x , beliefs are uniform on $[x, M]$. Write

$$F_x(z) = \begin{cases} 0, & z \leq x, \\ \frac{z-x}{M-x}, & x < z < M, \\ 1, & z \geq M. \end{cases}$$

If tactic τ is repressed in period 2 by factor $\gamma \in [0, 1]$, its participation cutoff is

$$\hat{c}_{2,\gamma}^\tau(x) = \int_0^1 F_x(\gamma A_\tau(\ell)) d\ell.$$

Let $L(c, u)$ denote turnout generated by participation cutoff c and common participation-cost realization u . By Lemma 2, $L(c, u)$ is weakly increasing in c for every u . Define the movement's actual period-2 success probability under the repressed tactic by

$$S_{2,\gamma}^\tau(x) = \mathbb{E}_u \left[F_x \left(\gamma A_\tau \left(L(\hat{c}_{2,\gamma}^\tau(x), u) \right) \right) \right],$$

and write $S_2^\tau(x) := S_{2,1}^\tau(x)$.

C.1 Responsive repression

Lemma A4 (Success under tactic-specific repression). *Fix x and τ . The function $S_{2,\gamma}^\tau(x)$ is continuous and weakly increasing in γ . It is strictly increasing over any interval on which it is positive.*

Proof. Take $\gamma_2 > \gamma_1$. Since F_x is weakly increasing,

$$\hat{c}_{2,\gamma_2}^\tau(x) \geq \hat{c}_{2,\gamma_1}^\tau(x).$$

Couple the two repression levels using the same realization u of the common participation-cost shock. Monotonicity of turnout in the participation cutoff gives

$$L(\hat{c}_{2,\gamma_2}^\tau(x), u) \geq L(\hat{c}_{2,\gamma_1}^\tau(x), u).$$

Because A_τ is weakly increasing,

$$\begin{aligned} \gamma_2 A_\tau \left(L(\hat{c}_{2,\gamma_2}^\tau(x), u) \right) &\geq \gamma_2 A_\tau \left(L(\hat{c}_{2,\gamma_1}^\tau(x), u) \right) \\ &\geq \gamma_1 A_\tau \left(L(\hat{c}_{2,\gamma_1}^\tau(x), u) \right). \end{aligned}$$

Applying the increasing function F_x and averaging over u establishes weak monotonicity. If the lower-repression-level success probability is positive, a positive-measure set of common shocks generates power strictly above the posterior lower bound. On this set, increasing γ strictly increases realized power, and the posterior has positive density over the relevant interval, yielding strict monotonicity. Continuity follows from continuity of the cutoff and realized-power mappings together with dominated convergence. $\square$

The next result is the formal version of the substitution cap stated verbally in Proposition 5.

Proposition A2 (Formal substitution cap). *Fix a belief state x and tactics $\tau \neq \tau'$ satisfying*

$$S_2^\tau(x) > S_2^{\tau'}(x) > 0.$$

Suppose only tactic τ is repressed. Then there exists a unique threshold

$$\hat{\gamma}_\tau(x) \in (0, 1)$$

such that the organizer prefers τ for $\gamma > \hat{\gamma}_\tau(x)$ and τ' for $\gamma < \hat{\gamma}_\tau(x)$. Defining

$$V_2(x, \gamma) = \max \left\{ S_{2,\gamma}^\tau(x), S_2^{\tau'}(x) \right\},$$

we have

$$V_2(x, 1) - V_2(x, \gamma) \leq S_2^\tau(x) - S_2^{\tau'}(x)$$

for every $\gamma \in [0, 1]$, with equality for every $\gamma \leq \hat{\gamma}_\tau(x)$.

Proof. By Lemma A4, $S_{2,\gamma}^\tau(x)$ is continuous and strictly increasing wherever it takes the positive fallback value. At $\gamma = 1$,

$$S_{2,1}^\tau(x) > S_2^{\tau'}(x),$$

whereas at $\gamma = 0$,

$$S_{2,0}^\tau(x) = 0 < S_2^{\tau'}(x).$$

The intermediate value theorem therefore gives a unique $\hat{\gamma}_\tau(x)$ satisfying

$$S_{2,\hat{\gamma}_\tau(x)}^\tau(x) = S_2^{\tau'}(x).$$

Hence

$$V_2(x, \gamma) = \begin{cases} S_{2,\gamma}^\tau(x), & \gamma > \hat{\gamma}_\tau(x), \\ S_2^{\tau'}(x), & \gamma \leq \hat{\gamma}_\tau(x), \end{cases}$$

up to tie-breaking at equality, and the stated bound follows immediately. Proposition 5 is the verbal implication of this result. $\square$

The violence-aversion parameter λ generates an additional asymmetry at the substitution boundary.

Corollary A1 (Regime incentives at the substitution boundary). *Suppose repression costs are strictly increasing in intensity and the regime can target only one tactic in period 2.*

- (i) *If peace is initially preferred and violence is the viable fallback, no optimal crowd-control policy strictly overshoots the switching threshold to violence.*
- (ii) *If violence is initially preferred and peace is the viable fallback, then for sufficiently large λ there exists a counterinsurgency policy that induces peace and strictly dominates no repression.*

Proof. For part (i), let $\hat{\gamma}_p(x)$ be the switching threshold. For every $\gamma < \hat{\gamma}_p(x)$, the organizer has already abandoned peace, so its success probability is fixed at the unrepressed violent fallback value. Replacing γ by $\hat{\gamma}_p(x)$ weakly preserves that success probability while strictly lowering repression costs; depending on tie-breaking, it also weakly lowers the violence cost. Thus every strict overshoot is dominated.

For part (ii), choose any $\gamma < \hat{\gamma}_v(x)$ that induces peace. Relative to no repression, the induced-peace policy replaces the initially preferred violent success probability $S_2^v(x)$ with the lower fallback value $S_2^p(x)$, incurs finite repression cost $\kappa_v(1 - \gamma)$, and avoids the direct violence cost λ . Hence it strictly dominates no repression whenever

$$\lambda > \kappa_v(1 - \gamma) - (S_2^v(x) - S_2^p(x)).$$

The right-hand side is finite, so the claim follows. $\square$

C.2 Temporary repression and the belief refund

The main-text Proposition 6 suppresses two auxiliary objects. For a fixed continuation tactic τ , define its unconditional contribution to lifetime success after a first-period failure at power x by

$$C_\rho^\tau(x) := \frac{M - x}{M} S_2^\tau(x).$$

Let

$$C_\rho(x) := \max_{\tau \in \{p, v\}} C_\rho^\tau(x)$$

and define the movement's eventual success probability conditional on first-period realized power x by

$$H_\rho(x) = \frac{x}{M} + C_\rho(x).$$

Proposition A3 (Formal belief-refund decomposition). *For every $x' < x$,*

$$C_\rho(x') \geq C_\rho(x),$$

and therefore

$$H_\rho(x) - H_\rho(x') = \frac{x - x'}{M} - [C_\rho(x') - C_\rho(x)].$$

Thus lowering realized first-period power weakly improves continuation prospects and offsets part of the direct loss in first-period success. Moreover, for

$$0 \leq x' < x \leq \min\{\alpha, c_\rho(\alpha, \beta; M)\},$$

we have

$$H_\rho(x') = H_\rho(x) \quad \text{if } \rho\beta = 0,$$

while

$$H_\rho(x') > H_\rho(x) \quad \text{if } \rho > 0 \text{ and } \beta > 0.$$

Proof. Fix a continuation tactic τ and consider $x' < x$. Under the lower posterior truncation point x' , citizens are weakly more optimistic, so

$$\hat{c}_2^\tau(x') \geq \hat{c}_2^\tau(x).$$

Using the same common shock u in both continuation games gives

$$L(\hat{c}_2^\tau(x'), u) \geq L(\hat{c}_2^\tau(x), u),$$

and hence

$$A_\tau(L(\hat{c}_2^\tau(x'), u)) \geq A_\tau(L(\hat{c}_2^\tau(x), u)).$$

Conditional on u , the continuation-success interval after failure at x is therefore contained in the corresponding interval after x' . Averaging over u yields

$$C_\rho^\tau(x') \geq C_\rho^\tau(x).$$

Taking the maximum over tactics preserves the inequality, establishing the decomposition.

For the full- and over-refund claims, let

$$z \leq \min\{\alpha, c_\rho(\alpha, \beta; M)\}.$$

Because $z \leq c_\rho$, violence is optimal in continuation. Because $z \leq \alpha$, even zero-turnout violence reaches the posterior lower bound. Its participation cutoff is therefore

$$\hat{c}_2^v(z) = \frac{\alpha + \beta/2 - z}{M - z}.$$

Using the turnout distribution from Lemma 2, the actual period-2 success probability is

$$S_2^v(z) = \rho(1 - \hat{c}_2^v(z)) \frac{\alpha - z}{M - z} + (1 - \rho)\hat{c}_2^v(z) + \rho\hat{c}_2^v(z) \frac{\alpha + \beta - z}{M - z}.$$

Substituting into lifetime success gives

$$H_\rho(z) = \frac{\alpha + (1 - \rho)\beta/2}{M} + \frac{\rho\beta(\alpha + \beta/2 - z)}{M(M - z)}.$$

Differentiating,

$$H'_\rho(z) = -\frac{\rho\beta(M - \alpha - \beta/2)}{M(M - z)^2}.$$

Since $M > 1 > \alpha + \beta$, the final factor is strictly positive. Thus $H'_\rho(z) = 0$ when $\rho\beta = 0$ and $H'_\rho(z) < 0$ when $\rho > 0$ and $\beta > 0$, proving the result. Proposition 6 follows immediately. $\square$

C.3 Persistence of violence

The no-period-2-repression assumption in Proposition 7 is a benchmark used to isolate the informational channel. Allowing the regime to re-optimize repression after the first failure would superimpose the responsive-substitution mechanism characterized above: in particular, second-period counterinsurgency could itself prevent the violent recurrence whose informational source we want to identify here.

Let γ scale only the period-1 violent technology. Under the prior, the period-1 participation cutoff when the organizer opens violently is

$$\hat{c}_{1,\gamma}^v = \int_0^1 \frac{\gamma(\alpha + \beta\ell)}{M} d\ell = \frac{\gamma(\alpha + \beta/2)}{M}.$$

Let X_γ denote realized first-period violent power under this policy, and let $c_\rho = c_\rho(\alpha, \beta; M)$ denote the continuation cutoff from Proposition 3.

Lemma A5 (Nested violent-recurrence events). *Consider two counterinsurgency policies $\gamma' < \gamma$ under which the organizer opens violently in both cases. Coupling them using the same common participation-cost shock and regime type gives*

$$X_{\gamma'} < X_\gamma.$$

Consequently, if violence recurs in period 2 under γ , then it also recurs under the stronger policy γ' .

Proof. Because $\gamma' < \gamma$,

$$\hat{c}_{1,\gamma'}^v < \hat{c}_{1,\gamma}^v.$$

Under the same common participation-cost shock,

$$\ell_{\gamma'} \leq \ell_\gamma,$$

so

$$X_{\gamma'} = \gamma'(\alpha + \beta\ell_{\gamma'}) < \gamma(\alpha + \beta\ell_\gamma) = X_\gamma.$$

Violence recurs under γ exactly when

$$\theta \geq X_\gamma \quad \text{and} \quad X_\gamma < c_\rho.$$

These inequalities imply

$$\theta \geq X_\gamma > X_{\gamma'} \quad \text{and} \quad X_{\gamma'} < X_\gamma < c_\rho,$$

which are sufficient for recurrence under γ' . □

Proof of Proposition 7. By Lemma A5, the event of a second violent episode is weakly larger under stronger counterinsurgency as long as both policies preserve a violent opening. Since violence occurs with certainty in period 1 on both branches, the expected number of violent periods is therefore weakly larger under the stronger policy.

If counterinsurgency instead induces a peaceful opening, violence can occur only in period 2. Its expected incidence is strictly below one because peaceful protest has a strictly positive probability of succeeding in period 1. Every policy that preserves a violent opening, by contrast, generates one violent period with certainty and therefore has expected violent incidence at least one. Any policy inducing a peaceful opening consequently produces strictly less expected violence. $\square$